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Introduction

Basic Topology

2.1 度量空间

Definition 2.1.1

令 $X \neq \emptyset$ 是一个非空集合. X 上的一个度量, 是指一个映射

$$\begin{aligned} d : X \times X &\rightarrow \mathbb{R} \\ (x, y) &\mapsto d(x, y) \end{aligned}$$

使得对于任意的 $x, y, z \in X$, 有

1. $d(x, y) \geq 0$, 且 $d(x, y) = 0 \iff x = y$
2. $d(x, y) = d(y, x)$
3. $d(x, z) \leq d(x, y) + d(y, z)$

Example 2.1.2

$X \neq \emptyset$, 定义 $d : X \times X \rightarrow \mathbb{R}$

$$d(x, y) = \begin{cases} 0, & x = y \\ 1, & x \neq y \end{cases}$$

是一个度量.

Proof. 1.2. 是显然的. 3. 当 $x = z$ 时是显然的. 若 $x \neq z$, 则要么 $x \neq y$, 要么 $z \neq y$, 从而

$$d(x, z) = 1 \leq d(x, y) + d(y, z)$$

□

2.2 Exercises

Problem 2.2.1 ► Convex Bodies

A convex body is a compact, convex subset $K \subseteq \mathbb{R}^n$ which contains 0 as an interior point and which is symmetric around the origin .

1. Prove that the following formula defines a norm on $\mathbb{R}^n$:

$$N(x) := \inf \left\{ \lambda > 0 : \frac{x}{\lambda} \in K \right\}$$

2. Prove that K is homeomorphic to $\mathbb{D}^n$ and that ∂K is homeomorphic to $\mathbb{S}^{n-1}$.

Note.

1. 若 K 是一个单位球. 则 $N(x) = \|x\|$.
2. 证明 $N(x)$ 是一个 norm, 需要证明: 1. 非负正定 2. 绝对一次齐次 3. 三角不等式.
3. 由于 φ 的形式足够简单, 可以通过直接构造逆映射证明双射.

Lemma 2.2.2

X is compact, Y is Hausdorff . If $\exists f : X \rightarrow Y$ is a continuous bijection, then X, Y are homeomorphic.

Proof. 1. It is obvious that $N(x) \geq 0$. If $N(x) = 0$, then

$$\inf \left\{ \lambda > 0 : \frac{x}{\lambda} \in K \right\} = 0$$

which means that for all $\varepsilon > 0$, $\frac{x}{\varepsilon} \in K$. Since K is compact, $\text{dist}(K, 0) < \infty$. If $x \neq 0$, then $\|x\| > 0$. We take $\varepsilon_0 < \frac{1}{\text{dist}(K, 0) \cdot \|x\|}$. Then $\left\| \frac{x}{\varepsilon_0} \right\| > \text{dist}(K, 0)$. But $\frac{x}{\varepsilon_0} \in K$, which is a contradiction. For the second part, let $x \in \mathbb{R}^n, k \in \mathbb{R} \setminus \{0\}$. For all $\lambda > N(x)$,

$$\frac{x}{\lambda} \in K \implies \frac{kx}{k\lambda} \in K$$

, which shows that $N(kx) \leq kN(x)$. Replace k by $\frac{1}{k}$ and replace x by kx , we have $N(kx) \geq kN(x)$. Thus $N(kx) = kN(x)$. For the triangle

inequality, we need to show that

$$\inf\left\{\lambda > 0 : \frac{x_1 + x_2}{\lambda} \in K\right\} \leq \inf\left\{\lambda > 0 : \frac{x_1}{\lambda} \in K\right\} + \inf\left\{\lambda > 0 : \frac{x_2}{\lambda} \in K\right\}$$

Let $\lambda_0 = N(x_1 + x_2)$. Only need to show that

$$\frac{x_1 + x_2}{N(x_1) + N(x_2)} \in K$$

For all $\delta > 0$, we have

$$\frac{x_1}{N(x_1) + \delta} \in K, \quad \frac{x_2}{N(x_2) + \delta} \in K$$

. Since K is convex, we have the convex combination

$$\frac{1}{N(x_1) + N(x_2) + 2\delta} \left((N(x_2) + \delta) \frac{x_2}{N(x_2) + \delta} + (N(x_1) + \delta) \frac{x_1}{N(x_1) + \delta} \right) \in K$$

That is

$$\frac{x_1 + x_2}{N(x_1) + N(x_2) + 2\delta} \in K$$

for all $\delta > 0$. Then

$$N(x_1 + x_2) \leq N(x_1) + N(x_2) + 2\delta, \quad \forall \delta > 0 \implies N(x_1 + x_2) \leq N(x_1) + N(x_2)$$

2. $x \in K^\circ \iff N(x) < 1$, $x \in \text{Ext}(K) \iff N(x) > 1$, $x \in \partial K \iff N(x) = 1$. We define a map $\varphi : \mathbb{R}^n \rightarrow \mathbb{R}^n$.

$$\varphi(x) := \begin{cases} N(x) \frac{x}{\|x\|}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

We have

$$\|\varphi(x)\| = N(x)$$

Then φ maps K° on to $(\mathbb{D}^n)^\circ$, and maps ∂K on to $\mathbb{S}^{n-1}$. To show φ is continuous, only need to show it is continuous at 0. In fact, $\|\lim_{x \rightarrow 0} \varphi(x)\| = \lim_{x \rightarrow 0} \|\varphi(x)\| = \lim_{x \rightarrow 0} N(x) = 0$ since N is a norm. We define $\psi : \mathbb{R}^n \rightarrow \mathbb{R}^n$:

$$\psi(x) := \begin{cases} \frac{\|x\|}{N(x)} x, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

Then $\varphi \circ \psi = \psi \circ \varphi = \text{Id}$. The same skills can be used to show ψ is continuous at 0 as well. Finally, by the lemma above, X, Y are homeomorphic.

□

Problem 2.2.3

Let X, X' be topological spaces and $x \in X, x' \in X'$ be base points. Prove that the wedge sum $X \vee X'$ is homeomorphic to the subspace:

$$(X \times \{x'\}) \cup (\{x\} \times X') \subseteq X \times X'$$

Proof. $X \vee X' = (X \amalg X') / \{x \sim x'\}$. Considering map $\varphi : X \times X' \rightarrow X \vee X'$

$$\varphi(x_1, x_2) = \begin{cases} x_1, & x_2 = x' \\ x_2, & x_1 = x \end{cases}$$

□

Problem 2.2.4

The torus $\mathbb{T}$ is the quotient of $[0, 1]^2$ by the equivalence relation generated by $(0, y) \sim (1, y)$ for all $x, y \in [0, 1]$. Prove that $\mathbb{T}$ is homeomorphic to :

1. The product $\mathbb{S}^1 \times \mathbb{S}^1$
2. The quotient of $\mathbb{R}^2$ under the action of the (discrete) group $\mathbb{Z}^2$ by translation.

Problem 2.2.5 ► Projective spaces

Let $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. The n -dimensional projective space $\mathbb{K}\mathbb{P}^n$ is the orbit space of $\mathbb{K}^{n+1} \setminus \{0\}$ under the action of $\mathbb{K}^*$ by rescaling.

1. Prove that $\mathbb{R}\mathbb{P}^n$ is homeomorphic to the orbit space $\mathbb{S}^n \setminus \{\pm 1\}$, where $\{\pm 1\}$ acts by multiplication on $\mathbb{S}^n \subseteq \mathbb{R}^{n+1}$.
2. Prove that $\mathbb{C}\mathbb{P}^n$ is homeomorphic to the orbit space $\mathbb{S}^{2n+1} \setminus \mathbb{S}^1$, where $\mathbb{S}^1 = \{z \in \mathbb{C} : |z| = 1\}$ acts on $\mathbb{S}^{2n+1} \subseteq \mathbb{C}^{n+1}$ by multiplication.
3. Prove that $\mathbb{R}\mathbb{P}^1$ is homeomorphic to $\mathbb{S}^1$ and that $\mathbb{C}\mathbb{P}^1$ is homeomorphic to $\mathbb{S}^2$.

Proof. 1. Define

$$\pi : \mathbb{R}^{n+1} \setminus \{0\} \rightarrow \mathbb{RP}^n, \quad \pi(x) = [x]$$

and define

$$f : \mathbb{S}^n \rightarrow \mathbb{RP}^n = \pi|_{\mathbb{S}^n}$$

Then f is continuous. Take any $p \in \mathbb{RP}^n$, there exists $v \in \mathbb{R}^{n+1} \setminus \{0\}$, such that $p = [v]$. Let $x = \frac{v}{\|v\|}$, then $x \in \mathbb{S}^n$,

$$f(x) = [x] = [v] = p$$

thus f is surjective. If

$$f(x) = f(y)$$

, then $[x] = [y]$, there exists $\lambda \in \mathbb{R}^*$, such that $x = \lambda y$. Since

$$\|x\| = \|\lambda y\| = |\lambda| \|y\| \implies |\lambda| = 1$$

, $\lambda = \pm 1$, $x = y$ or $x = -y$. Thus $\text{Fiber}_p(f) = \text{Orbit}_x(\{\pm 1\})$. The universal property of quotient topology gives a continuous bijection

$$\tilde{f} : \mathbb{S}^n \setminus \{\pm 1\} \rightarrow \mathbb{RP}^n$$

Since $\mathbb{S}^n \setminus \{\pm 1\}$ is compact, $\mathbb{RP}^n$ is Hausdorff, $\tilde{f}$ is a homeomorphism.

2. Similarly, let $g : \mathbb{S}^{2n+1} \rightarrow \mathbb{CP}^n$, $g(z) = [z]$. Take any $p = [w] \in \mathbb{CP}^n$, where $w \in \mathbb{C}^{n+1} \setminus \{0\}$. Let $z = \frac{w}{\|w\|}$. Then

$$g(z) = [z] = [w] = p$$

g is surjective. If

$$g(z_1) = g(z_2)$$

, then there exists $\lambda \in \mathbb{C}^*$, such that $z_1 = \lambda z_2$, then

$$\|z_1\| = |\lambda| \|z_2\| \implies |\lambda| = 1$$

which gives that $\lambda \in \mathbb{S}^1$. Thus $\text{Fiber}_p(f) \simeq \text{Orbit}_x(\mathbb{S}^1)$. Thus

$$\mathbb{CP}^n \simeq \mathbb{S}^{2n+1}/\mathbb{S}^1$$

3. (a) Only need to show that $\mathbb{S}^1 / \{\pm 1\} \simeq \mathbb{S}^1$,

$$h(z) = z^2$$

induces a continuous bijection $\tilde{h} : \mathbb{S}^1 \setminus \{\pm 1\} \rightarrow \mathbb{S}^1$. $\mathbb{S}^1 \setminus \{\pm 1\}$ is compact, $\mathbb{S}^1$ is Hausdorff, $\tilde{h}$ is a homeomorphism.

(b) Since $\mathbb{S}^3/\mathbb{S}^1 \simeq \mathbb{S}^2$.

□

Problem 2.2.6

The torus $\mathbb{T}$ is the quotient of $[0, 1]^2$ by the equivalence relation generated by $(x, 0) \sim (x, 1)$ and $(0, y) \sim (1, y)$ for all $x, y \in [0, 1]$. Prove that $\mathbb{T}$ is homeomorphic to:

1. The product $\mathbb{S}^1 \times \mathbb{S}^1$.
2. The quotient of $\mathbb{R}^2$ under the action of the (discrete) group $\mathbb{Z}^2$ by translations.

Proof. 1. Let

$$f(x, y) := (e^{2\pi xi}, e^{2\pi yi})$$

Then

$$f(x, 0) = (e^{2\pi ix}, 1) = f(x, 1)$$

$$f(0, y) = (1, e^{2\pi yi}) = f(1, y)$$

Thus f induces a map $\bar{f}$ on $\mathbb{T}$ such that $\bar{f}([(x, y)]) = (e^{2\pi xi}, e^{2\pi yi})$, which is a continuous function. It is not hard to see that $\bar{f} : \mathbb{T} \rightarrow \mathbb{S}^1 \times \mathbb{S}^1$ is a bijection. As a continuous image of $[0, 1]^2$, $\mathbb{T}$ is compact. Take any closed subset V of $\mathbb{T}$, V is compact. $\bar{f}(V)$ is compact as the continuous image of V . Since $\mathbb{S}^1 \times \mathbb{S}^1$ is Hausdorff, $\bar{f}(V)$ is closed. Thus $\bar{f}$ is a continuous and closed bijection, thus is a homeomorphism.

2. Let

$$f(x, y) = (x, y) / \mathbb{Z}^2$$

Then Since $(x, 0) - (x, 1) \in \mathbb{Z}^2, (0, y) - (1, y) \in \mathbb{Z}^2$. f induces a continuous map $\bar{f}$ on from $\mathbb{T}$ to $\mathbb{S}^1 \times \mathbb{S}^1$. It is clear that f maps onto $\mathbb{R}^2/\mathbb{Z}^2$. If $f(x_1, y_1) = f(x_2, y_2)$, then

$$(x_1 - x_2, y_1 - y_2) \in \mathbb{Z}^2 \implies x_1 = x_2 \text{ or } \{x_1, x_2\} = \{0, 1\}, \text{ and } y_1 = y_2 \text{ or } \{y_1, y_2\} = \{0, 1\}$$

thus $(x_1, y_1) \sim (x_2, y_2)$. Again since $\mathbb{T}$ is compact, $\mathbb{R}^2/\mathbb{Z}^2$ is Hausdorff, $\bar{f}$ is a homomorphism. □

Problem 2.2.7

Consider the orthogonal group $O_{n-1}(\mathbb{R})$ as the subgroup of $O_n(\mathbb{R})$ of matrices of the form:

$$\begin{pmatrix} 1 & 0 \\ 0 & A \end{pmatrix} \in O_n(\mathbb{R}), A \in O_{n-1}(\mathbb{R}).$$

Let $O_{n-1}(\mathbb{R})$ act on $O_n(\mathbb{R})$ by left multiplication. Prove that the orbit space $O_n(\mathbb{R})/O_{n-1}(\mathbb{R})$ is homeomorphic to $\mathbb{S}^{n-1}$.

Proof sketch.

将 $O_n(\mathbb{R})$ 刻画为 $\mathbb{S}^{n-1}$ 上的群作用. 此时 $A \sim \begin{pmatrix} 1 & 0 \\ 0 & A \end{pmatrix}$ 就是北极点的稳定子群. 一但作用是传递的, 由流形上的轨道-稳定化子定理, 就有

$$O_n(\mathbb{R})/O_{n-1}(\mathbb{R}) \simeq \mathbb{S}^{n-1}$$

Fundamental Group

3.1 Exercises

Problem 3.1.1 ► Mobius band

Let M be the Mobius band, that is, the quotient of $[0, 1]^2$ under the equivalence relation generated by $(x, 0) \sim (1 - x, 1)$ for all $x \in [0, 1]$. Prove that M is homotopy equivalent to $\mathbb{S}^1$.

Proof. Construct

$$\begin{aligned} f : [0, 1]^2 \times [0, 1] &\rightarrow \mathbb{S}^1 \\ (x, y, t) &\mapsto \left(\frac{1}{2}t + (1 - t)x, y \right) \end{aligned}$$

Denote $f_t(x, y) = f(x, y, t)$, then

$$f_0(x, y) = (x, y), \quad f_1(x, y) = \left(\frac{1}{2}, y \right), \quad f_t\left(\frac{1}{2}, y\right) \equiv \left(\frac{1}{2}, y\right)$$

Which is a SDR to $\mathbb{S}^1$. □

Problem 3.1.2

1. Prove that the inclusion $O_n(\mathbb{R}) \rightarrow GL_n(\mathbb{R})$ is a homotopy equivalence. (Use the Gram-Schmidt orthonormalization procedure to construct the reverse map).
2. Among the following matrix groups, determine which ones are compact, and determine their π_0 :

$$GL_n(\mathbb{C}), GL_n(\mathbb{R}), O_n(\mathbb{R}), SO_n(\mathbb{R}), U_n(\mathbb{C}), SU_n(\mathbb{C}).$$

Proof. Take $A \in \mathrm{GL}(n, \mathbb{R})$. Let $A = (v_1, \dots, v_n)$. Let

$$\begin{aligned} e_1 &= \frac{v_1}{\|v_1\|} \\ e_2 &= \frac{v_2 - \langle v_2, e_1 \rangle e_1}{\|v_2 - \langle v_2, e_1 \rangle e_1\|} \\ &\dots \end{aligned}$$

$\det Q \det(t \mathrm{Id} + (1 - t) R) \neq 0$. Let $Q = (e_1, \dots, e_n)$. $A = QR$ for some upper triangular matrix R . Let

$$\begin{aligned} f_t &: \mathrm{GL}_n(\mathbb{R}) \rightarrow \mathrm{GL}_n(\mathbb{R}) \\ A &\mapsto Q(t \mathrm{Id} + (1 - t) R) \end{aligned}$$

which is a SDR from A to Q . □

Covering Space

4.1 Basic Definitions

Definition 4.1.1 ► Covering Space

Let $p : \bar{X} \rightarrow X$ be a surjective map.

- We say an open subset V of X is **evenly covered** by p , if $p^{-1}(V)$ is a disjoint union of open subsets of $\bar{X}$:

$$p^{-1}(V) = \coprod_i U_i$$

where, each U_i is mapped homeomorphically onto V by p .

- In this case, we shall refer to each U_i as a **sheet** for p over V .
- If the space X can be covered by open subsets which are evenly covered by p , then we say that p is a **covering projection**; the space $\bar{X}$ is called a covering space of X .

Theorem 4.1.2

Let $p : \bar{X} \rightarrow X$ be a continuous map, where X is locally path connected.

1. The map p is a covering space iff for each component $\bar{C}$ of $\bar{X}$, the restriction map $p : p^{-1}(C) \rightarrow C$ is a covering projection.
2. If p is a covering projection then for each component $\bar{C}$ of $\bar{X}$, the map $p : \bar{C} \rightarrow p(\bar{C})$ is a covering projection and $p(\bar{C})$ is a component of X .

Remark.

Thus, we always assume that a covering space is locally path connected and connected. For the following content, unless otherwise specified, we will all adopt this assumption.

Proposition 4.1.3

Let $p : \bar{X} \rightarrow X$ be a covering projection. If X is Hausdorff then $\bar{X}$ is Hausdorff. Further, if p is finite-to-one, then show that if $\bar{X}$ is Hausdorff then X is Hausdorff.

Note.

I haven't finish the proof of the second part yet. Actually, if a covering projection is proper it is also closed. Once we have this conclusion we can then finish the proof easily.

Proof. 1. Take two different points $\bar{x}, \bar{y} \in \bar{X}$. Denote $x := p(\bar{x}), y := p(\bar{y})$.

- (a) If $x \neq y$: Since X is Hausdorff, there exists two open sets U, V such that $x \in U, y \in V, U \cap V = \emptyset$. We have $p^{-1}(U), p^{-1}(V)$ are two open subsets of $\bar{X}$ with $p^{-1}(U) \cap p^{-1}(V) = \emptyset$. Since $\bar{x} \in p^{-1}(U), \bar{y} \in p^{-1}(V)$, it shows that $\bar{X}$ is Hausdorff.
- (b) If $x = y$: Let W be a open neighbourhood of $x = y$. We write $p^{-1}(W) = \coprod_i W_i$. Since $\bar{x}, \bar{y} \in \coprod_i W_i$ and any two W_i are disjoint, there are two different W_{i_1}, W_{i_2} such that $\bar{x} \in W_{i_1}, \bar{y} \in W_{i_2}$.

The above shows that $\bar{X}$ is Hausdorff.

- 2. Take two different points $x, y \in X$. Then $p^{-1}(x), p^{-1}(y)$ are two finite sets of $\bar{X}$ with empty intersection. We write

$$p^{-1}(x) = \{\bar{x}_i\}_{i=1}^n, \quad p^{-1}(y) = \{\bar{y}_i\}_{i=1}^n$$

For each $i = 1, \dots, n$, let $U_i \ni \bar{x}_i, V_i \ni \bar{y}_i, U_i \cap V_i = \emptyset$. Then $p(\bigcup_i U_i) \cap p(\bigcup_i V_i) = \emptyset, x \in p(\bigcup_i U_i), y \in p(\bigcup_i V_i)$.

□

Basic Homology Algebra

5.1 Basic definition

Definition 5.1.1

Consider a direct sum

$$C_{\bullet} := C_* := \bigoplus_{n \in \mathbb{Z}} C_n$$

of R -modules^a. Often we call C_* a **graded-module** with its n^{th} **graded component** C_n . Members of C_n are also called **homogeneous elements** of C_* of degree n .

1. A **R -module homomorphism** $f : C_* \rightarrow C'_*$ is called a graded homomorphism if there exists d such that $f(C_r) \subseteq C'_{r+d}$ for all r . We then call d the degree of f . We shall denote $f|_{C_r}$ by f_r , and often we may simply write f itself for f_r provided that there is no confusion.
2. By a **chain complex** (C_*, ∂) of R -modules, we mean a graded R -module C_* , together with an endomorphism $\partial := \partial_* : C_* \rightarrow C_*$ of degree -1 with the property $\partial \circ \partial = 0$. The endomorphism ∂ is called the **differential** or the **boundry map** of the chain complex. Often we shall not mention the ∂ at all and merely say C_* is a chain complex.
3. If C and C_* are two chain complexes then by a **chain map** $f = f_* : C_* \rightarrow C'_*$ we mean a graded module homomorphism of degree 0 that commutes with the corresponding differentials.

^ais also a R -module

Remark.

1. The direct sum C_* is also an R -module.
2. Observe that ∂ consists of a sequence $\{\partial_n : C_n \rightarrow C_{n-1}\}$ of R -module homomorphisms such that $\partial_n \circ \partial_{n-1} = 0$ for all n .
3. f consists of a sequence $\{f_n : C_n \rightarrow C'_n\}$ of R -module homomorphisms such that $\partial'_n \circ f_n = f_{n-1} \circ \partial_n$ for all n . Expressed with an

diagram, that is

$$\begin{array}{ccc} C_n & \xrightarrow{\partial_n} & C_{n-1} \\ \downarrow f_n & & \downarrow f_{n-1} \\ C'_n & \xrightarrow{\partial'_n} & C'_{n-1} \end{array}$$

Proposition 5.1.2

There is a category of chain complexes of R -modules and chain maps. We shall denote this category by $\mathbf{Ch}_R$.

Proof. The objects are chain complexes. The morphisms are chain maps. For composition of morphisms, consider two chain maps $f : C_* \rightarrow C'_*$, $g : C'_* \rightarrow C''_*$. Only need to show that the following diagram commutes

$$\begin{array}{ccc} C_n & \xrightarrow{\partial_n} & C_{n-1} \\ \downarrow f_n & & \downarrow f_{n-1} \\ C'_n & \xrightarrow{\partial'_n} & C'_{n-1} \\ \downarrow f'_n & & \downarrow f'_{n-1} \\ C''_n & \xrightarrow{\partial''_n} & C''_{n-1} \end{array}$$

which is obvious. Finally, the existence of the identity map is also obvious. □

Definition 5.1.3 ▶ Direct Sum of Chain Complexes

Define the direct sum of a family of chain complexes $\{(C^\alpha, \partial^\alpha)\}_{\alpha \in \Lambda}$ as the chain complexes $(C, \partial) := (\bigoplus_\alpha C^\alpha, \bigoplus_\alpha \partial^\alpha)$, where the n^{th} graded component of C is

$$C_n = (C_n^\alpha)_{\alpha \in \Lambda},$$

∂ is defined as

$$\partial((c^\alpha)_{\alpha \in \Lambda}) := (\partial^\alpha(c^\alpha))_{\alpha \in \Lambda}$$

5.2 Exact Sequence

Definition 5.2.1 ► Exact Sequence

1. A sequence of R -modules

$$M' \xrightarrow{\alpha} M \xrightarrow{\beta} M''$$

is said to be **exact** at M if $\ker \beta = \operatorname{Im} \alpha$.

2. A sequence

$$\cdots \longrightarrow M_{n-1} \longrightarrow M_n \longrightarrow M_{n+1} \longrightarrow \cdots$$

is said to be exact if it is exact at each M_n .

3. By a short exact sequence we mean an exact sequence of the form

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

Definition 5.2.2 ► Exact Sequence of Chain Complexes

A sequence of chain complexes and chain maps

$$0 \longrightarrow C' \xrightarrow{f} C \xrightarrow{g} C'' \longrightarrow 0$$

is said to be exact if for each n the corresponding sequence of modules

$$0 \longrightarrow C'_n \xrightarrow{f_n} C_n \xrightarrow{g_n} C''_n \longrightarrow 0$$

is exact.

Lemma 5.2.3 ► Snake lemma

Given a commutative diagram of R -module homomorphisms: where the two horizontal sequences are exact, there exists a R -module homomorphism

$$\begin{array}{ccccccc} M' & \xrightarrow{\alpha} & M & \xrightarrow{\beta} & M'' & \longrightarrow & 0 \\ \downarrow f' & & \downarrow f & & \downarrow f'' & & \\ 0 & \longrightarrow & N' & \xrightarrow{\alpha'} & N & \xrightarrow{\beta'} & N'' \end{array}$$

$\delta : \operatorname{Ker} f'' \longrightarrow \operatorname{Coker} f'$, called the *connecting homomorphism* such that the sequence

$$\operatorname{Ker} f' \xrightarrow{\bar{\alpha}} \operatorname{Ker} f \xrightarrow{\bar{\beta}} \operatorname{Ker} f'' \xrightarrow{\delta} \operatorname{Coker} f' \xrightarrow{\bar{\alpha}'} \operatorname{Coker} f \xrightarrow{\bar{\beta}'} \operatorname{Coker} f''$$

is exact. Moreover, the **connecting homomorphism** δ has the naturality properties, so that the above assignment of a ‘snake’ to the corresponding ‘six-term’ exact sequence of modules defines a covariant functor.

Note.

最关键的信息是 **connecting homomorphism** δ . 构造的过程就是借助 β 的是 surjective, α' 是 injective, 来调整使得 β 和 α' 对应的箭头是可以反转的. 事实上在应用时, 这个 connecting homomorphism 的具体构造往往比它的存在性更重要.

Proof. to be finished We need to show

1. $\bar{\alpha}, \bar{\beta}$ are well defined, where $\bar{\alpha} := \alpha|_{\ker f'}, \bar{\beta} := \beta|_{\ker f}$
2. $\text{Im } \bar{\alpha} = \ker \bar{\beta}$.
3. $\bar{\alpha}', \bar{\beta}'$ are well defined, where $\bar{\alpha}'([x]) := [\alpha'(x)], \bar{\beta}'([y]) := [\beta'(y)]$
4. $\text{Im } \bar{\alpha}' = \ker \bar{\beta}'$
5. Construct a δ .
6. $\text{Im } \bar{\beta} = \ker \delta$
7. $\text{Im } \delta = \ker \bar{\alpha}'$

We prove the above points below.

1. By commutative, $f \circ \alpha(\ker f') = \alpha' \circ f'(\ker f') = 0$, which shows that $\alpha(\ker f') \subseteq \ker f$. Thus $\bar{\alpha}$ is well defined. $\bar{\beta}$'s is similar.
- 2.

$$\text{Im } \bar{\alpha} = \alpha(\ker f') \subseteq \text{Im } \alpha = \ker \beta \implies \beta(\text{Im } \bar{\alpha}) = 0 \implies \text{Im } \bar{\alpha} \subseteq \ker \bar{\beta}$$

The other side,

$$\ker \bar{\beta} = \ker \beta \cap \ker f = \text{Im } \alpha \cap \ker f$$

Take $y \in \ker \bar{\beta}$, there exists $x \in M'$ such that $y = \alpha(x)$ and $f(y) = f(\alpha(x)) = 0$. Since $f \circ \alpha = \alpha' \circ f'$, $\alpha' \circ f'(x) = 0$. Since α' is injective, $x \in \ker f'$. Thus $y \in \text{Im } \bar{\alpha}$.

3. For $\bar{\alpha}'$, we need to show that for $x_1, x_2 \in N'$ such that $x_1 - x_2 \in \text{Im } f'$, $\alpha'(x_1) - \alpha'(x_2) \in \text{Im } f$. It is true by commutative

$$\alpha'(x_1) - \alpha'(x_2) = \alpha'(x_1 - x_2) \in \text{Im } (\alpha' \circ f') = \text{Im } (f \circ \alpha) \subseteq \text{Im } f$$

$\bar{\beta}'$'s is similar.

4. Take $[y]_{\sim \text{Im } f} \in \text{Im } \bar{\alpha}'$, then there exists $[x]_{\sim \text{Im } f'} \in \text{Coker } f'$, such that $\alpha'([x]_{\sim \text{Im } f'}) = [y]_{\sim \text{Im } f}$. Then $\alpha'(x) - y \in \text{Im } f$, there exists $z \in M$, such that

$$\alpha'(x) - y = f(z)$$

By commutative and exactness: $\ker \beta' = \text{Im } \alpha'$,

$$\beta'(y) = \beta'(\alpha'(x) - f(z)) = 0 - \beta' \circ f(z) = -f'' \circ \beta(z) \in \text{Im } f''$$

, which shows that $\bar{\beta}'([y]_{\sim \text{Im } f}) = [0]_{\sim \text{Im } f''}$, $\text{Im } \bar{\alpha}' \subseteq \ker \bar{\beta}'$.

Then, take $[y]_{\sim \text{Im } f} \in \ker \bar{\beta}'$. Then $y \in \ker \beta' = \text{Im } \alpha'$. There exists $x \in N'$ such that $y = \alpha'(x)$. Then $[y]_{\sim \text{Im } f} = \alpha'([x]_{\sim \text{Im } f'})$, $[y]_{\sim \text{Im } f} \in \text{Im } \bar{\alpha}'$, $\ker \bar{\beta}' \subseteq \text{Im } \bar{\alpha}'$.

5. Take $z \in \ker f''$. Since β is surjective, there exists $y \in M$ such that $\beta(y) = z$. Once we show that $f(y) \in \text{Im } \alpha'$, we can define

$$\delta(z) = [(\alpha')^{-1}(f(y))]_{\sim \text{Im } f'}$$

provided that this definition is well defined. By commutative

$$\beta'(f(y)) = f''(\beta(y)) = f''(z) = 0.$$

Hence $f(y) \in \ker \beta' = \text{Im } \alpha'$. Finally, to show that it is well defined, we take $z_1, z_2 \in \ker f''$ such that $\beta(y_1) = \beta(y_2) = z$. We need to show that $(\alpha')^{-1}(f(y_1 - y_2)) \in \text{Im } f'$. Since $\beta(y_1) - \beta(y_2) = 0$, $\beta(y_1 - y_2) = 0$, $y_1 - y_2 \in \ker \beta = \text{Im } \alpha$. Suppose $\alpha(x) = y_1 - y_2$. Then

$$(\alpha')^{-1}(f(y_1 - y_2)) = (\alpha')^{-1}(f\alpha(x)) = (\alpha')^{-1}(\alpha' \circ f')(x) = f'(x) \in \text{Im } f'$$

, which completes the proof. □

Corollary 5.2.4

Consider the following commutative diagram of R -modules and R -linear maps in which the two rows are exact. If f_1 and f_3 are isomorphisms then so is f_2 .

$$\begin{array}{ccccccc}
0 & \longrightarrow & M_1 & \xrightarrow{\alpha_1} & M_2 & \xrightarrow{\alpha_2} & M_3 \longrightarrow 0 \\
& & \downarrow f_1 & & \downarrow f_2 & & \downarrow f_3 \\
0 & \longrightarrow & N_1 & \xrightarrow{\beta_1} & N_2 & \xrightarrow{\beta_2} & N_3 \longrightarrow 0
\end{array}$$

Proof. By snake lemma, we have

$$0 \xrightarrow{\alpha_1} \ker f_2 \xrightarrow{\alpha_2} 0 \xrightarrow{\delta} 0 \xrightarrow{\beta_1} \text{Coker } f_2 \xrightarrow{\beta_2} 0$$

is exact. Then we have

$$\begin{aligned}
0 &= \ker \alpha_2 = \text{Im } \alpha_1 = 0 \\
&= \ker f_2
\end{aligned}$$

which shows that $\ker f_2 = 0$. Similarly, $\text{Coker } f_2 = 0$. The above shows that f_2 is an isomorphism. $\square$

Corollary 5.2.5 ▶ Four lemma

Consider the following commutative diagram of R modules and R -linear maps in which the two rows are exact. Suppose that f_1 is surjective and f_4 is injective. Then

- (i) f_2 is injective $\implies f_3$ is injective.
- (ii) f_3 is surjective $\implies f_2$ is surjective.

$$\begin{array}{ccccccc}
M_1 & \xrightarrow{\alpha_1} & M_2 & \xrightarrow{\alpha_2} & M_3 & \xrightarrow{\alpha_3} & M_4 \\
\downarrow f_1 & & \downarrow f_2 & & \downarrow f_3 & & \downarrow f_4 \\
N_1 & \xrightarrow{\beta_1} & N_2 & \xrightarrow{\beta_2} & N_3 & \xrightarrow{\beta_3} & N_4
\end{array}$$

Proof sketch.

如果 f_3 被两个单射 f_2, f_4 夹在中间, 得益于 f_1 是满射, 可以将右三列保持单射性地调整为一条 snake, 从而导出 f_3 是单射.

类似地, 如果 f_2 被两个满射 f_1, f_3 夹在中间, 得益于 f_4 是单射, 左三列可以保持满射性地调整为一条 snake, 导出 f_2 是满射.

Proof. 1. We can construct a snake

$$\begin{array}{ccccccc}
 M_2/\ker \alpha_2 & \xrightarrow{\alpha_2} & M_3 & \xrightarrow{\alpha_3} & \operatorname{Im} \alpha_3 & \longrightarrow & 0 \\
 \downarrow \bar{f}_2 & & \downarrow f_3 & & \downarrow f_4|_{\operatorname{Im} \alpha_3} & & \\
 0 \longrightarrow & N_2/\ker \beta_2 & \xrightarrow{\beta_2} & N_3 & \xrightarrow{\beta_3} & N_4 &
 \end{array}$$

By snake lemma, the following sequence is exact

$$\ker \bar{f}_2 \rightarrow \ker f_3 \rightarrow \ker f_4|_{\operatorname{Im} \alpha_3}$$

Since f_4 is injective, $f_4|_{\operatorname{Im} \alpha_3}$ is as well. Furthermore, by applying the snake lemma to the following diagram

$$\begin{array}{ccccccc}
 M_1 & \xrightarrow{\alpha_1} & M_2 & \xrightarrow{p} & M_2/\ker \alpha_2 & \longrightarrow & 0 \\
 \bar{f}_1 \downarrow & & \downarrow f_2 & & \downarrow \bar{f}_2 & & \\
 0 \longrightarrow & N_1/\ker \beta_1 & \xrightarrow{\bar{\beta}_1} & N_2 & \xrightarrow{p} & N_2/\ker \beta_2 &
 \end{array}$$

We get

$$0 = \ker f_2 \rightarrow \ker \bar{f}_2 \xrightarrow{\delta} \operatorname{Coker} \bar{f}_1$$

is a exact suquence, where $\operatorname{Coker} \bar{f}_1 = 0$ since f_1 is surjective and $\bar{f}_1$ is as well. It shows that $\ker \bar{f}_2 = 0$. Finally, we have the following sequence is exact

$$0 = \ker \bar{f}_2 \rightarrow \ker f_3 \rightarrow \ker f_4|_{\operatorname{Im} \alpha_3} = 0$$

It follows that $\ker f_3 = 0$, f_3 is injective.

2. Another snake we can construct is

$$\begin{array}{ccccccc}
 M_1 & \xrightarrow{\alpha_1} & M_2 & \xrightarrow{\alpha_2} & \operatorname{Im} \alpha_2 = \ker \alpha_3 & \longrightarrow & 0 \\
 \downarrow \bar{f}_1 & & \downarrow f_2 & & \downarrow f_3|_{\operatorname{Im} \alpha_2} & & \\
 0 \longrightarrow & N_1/\ker \beta_1 & \xrightarrow{\bar{\beta}_1} & N_2 & \xrightarrow{\beta_2} & \operatorname{Im} \beta_2 = \ker \beta_3 &
 \end{array}$$

Then by snake lemma ,the following sequence is exact

$$\ker \bar{f}_1 \rightarrow \ker f_2 \rightarrow \ker f_3|_{\operatorname{Im} \alpha_2} \rightarrow \operatorname{Coker} \bar{f}_1 \rightarrow \operatorname{Coker} f_2 \rightarrow \operatorname{Coker} f_3|_{\operatorname{Im} \alpha_2}$$

Furthermore, by applying the snake lemma to the following diagram

$$\begin{array}{ccccccccc}
 0 & \longrightarrow & \text{Ker } \alpha_3 & \xrightarrow{\iota_M} & M_3 & \xrightarrow{\alpha_3} & \text{Im } \alpha_3 & \longrightarrow & 0 \\
 & & \downarrow f_3|_{\text{Im } \alpha_2} & & \downarrow f_3 & & \downarrow f_4|_{\text{Im } \alpha_3} & & \\
 0 & \longrightarrow & \text{Ker } \beta_3 & \xrightarrow{\iota_N} & N_3 & \xrightarrow{\beta_3} & \text{Im } \beta_3 & \longrightarrow & 0
 \end{array}$$

We get

$$0 = \ker f_4|_{\text{Im } \alpha_3} \xrightarrow{\delta} \text{Coker}(f_3|_{\text{Im } \alpha_2}) \rightarrow \text{Coker } f_3 = 0$$

which shows that $\text{Coker}(f_3|_{\text{Im } \alpha_2}) = 0$. Finally, we have the following sequence exact

$$0 = \text{Coker } \bar{f}_1 \rightarrow \text{Coker } f_2 \rightarrow \text{Coker } f_3|_{\text{Im } \alpha_2} = 0$$

Thus $\text{Coker } f_2 = 0$, f_2 is surjective.

□

Corollary 5.2.6 ▶ Five lemma

In the following diagram of R -modules, the two rows are given to be exact. If f_1, f_2, f_4 and f_5 are isomorphisms then f_3 is also an isomorphism.

$$\begin{array}{ccccccccc}
 M_1 & \longrightarrow & M_2 & \longrightarrow & M_3 & \longrightarrow & M_4 & \longrightarrow & M_5 \\
 \downarrow f_1 & & \downarrow f_2 & & \downarrow f_3 & & \downarrow f_4 & & \downarrow f_5 \\
 M'_1 & \longrightarrow & M'_2 & \longrightarrow & M'_3 & \longrightarrow & M'_4 & \longrightarrow & M'_5
 \end{array}$$

Proof. By applying the Four lemma to the left four columns, we get f_3 is injective. And by applying the Four lemma to the right four columns, we get f_3 is surjective.

□

5.3 Homology

Definition 5.3.1

Given a chain complex C_* , define the **homology group** of C_* to be the graded R -module

$$H_*(C_*) := \bigoplus_{n \in \mathbb{Z}} H_n(C_*)$$

by taking

$$H_n(C_*) := \text{Ker } \partial_n / \text{Im } \partial_{n+1}, \quad \forall n \in \mathbb{Z}.$$

Proposition 5.3.2

If $f : C_* \rightarrow C'_*$ is a chain map then f induces a graded homomorphism

$$H_*(f) : H_*(C_*) \rightarrow H_*(C'_*)$$

In addition, this has the naturality property, viz.,

1. $H_*(\text{Id}) = \text{Id}$
2. If g is another chain map such that $f \circ g$ is defined, then

$$H_*(f \circ g) = H_*(f) \circ H_*(g)$$

Thus, H_* is a **covariant functor** from the category of chain complexes to the category of graded modules.

Theorem 5.3.3

The homology of a direct sum of chain complexes is isomorphic to the direct sum of the homology of chain complexes.

Proof.

$$H_n(\bigoplus_{\alpha} C_*^{\alpha}) = \text{ker}(\bigoplus_{\alpha} \partial_n^{\alpha}) / \text{Im}(\bigoplus_{\alpha} \partial_{n+1}^{\alpha})$$

$$\bigoplus_{\alpha} H_n(C_*^{\alpha}) = \bigoplus_{\alpha} (\text{ker } \partial_n^{\alpha} / \text{Im } \partial_{n+1}^{\alpha})$$

There are obvious isomorphisms :

$$\text{ker}(\bigoplus_{\alpha} \partial_n^{\alpha}) \simeq \bigoplus_{\alpha} \text{ker } \partial_n^{\alpha}, \quad \text{Im}(\bigoplus_{\alpha} \partial_{n+1}^{\alpha}) \simeq \bigoplus_{\alpha} \text{Im } \partial_{n+1}^{\alpha}$$

A standard result in algebra is that if K_α is a submodule of M_α , then

$$\frac{\bigoplus_\alpha M_\alpha}{\bigoplus_\alpha K_\alpha} \cong \bigoplus_\alpha \frac{M_\alpha}{K_\alpha}$$

. Hence , we have

$$\bigoplus_\alpha \ker \partial_n^\alpha / \bigoplus_\alpha \operatorname{Im} \partial_{n+1}^\alpha \simeq \bigoplus_\alpha (\ker \partial_n^\alpha / \operatorname{Im} \partial_{n+1}^\alpha)$$

□

Theorem 5.3.4

Given a short exact sequence of chain complexes

$$0 \rightarrow C'_* \xrightarrow{\alpha} C_* \xrightarrow{\beta} C''_* \rightarrow 0$$

there is a functorial long exact sequence of homology groups

$$\rightarrow H_n(C'_*) \xrightarrow{H_n(\alpha)} H_n(C_*) \xrightarrow{H_n(\beta)} H_n(C''_*) \xrightarrow{\delta_n} H_{n-1}(C'_*) \xrightarrow{H_{n-1}(\alpha)} H_{n-1}(C_*) \rightarrow$$

Proof sketch.

Consider the diagram

$$\begin{array}{ccccccc} C'_n / \operatorname{Im} \partial'_{n+1} & \xrightarrow{\bar{\alpha}_n} & C_n / \operatorname{Im} \partial_{n+1} & \xrightarrow{\bar{\beta}_n} & C''_n / \operatorname{Im} \partial''_{n+1} & \longrightarrow & 0 \\ \downarrow \partial'_n & & \downarrow \partial_n & & \downarrow \partial''_{n+1} & & \\ 0 \longrightarrow & \operatorname{Ker} \partial'_{n-1} & \xrightarrow{\alpha'_{n-1}} & \operatorname{Ker} \partial_{n-1} & \xrightarrow{\beta'_{n-1}} & \operatorname{Ker} \partial''_{n-1} & \end{array}$$

Singular Homology

6.1 The Construction

Definition 6.1.1

Let X be any topological space and $n \geq 0$ be an integer. By a **singular n -simplex** in X , we mean a continuous map $\sigma : |\Delta_n| \rightarrow X$. By a **singular n -chain** in X , we mean a finite sum $\sum_i n_i \sigma_i$, where $n_i \in \mathbb{Z}$ and σ_i are singular n -simplices in X . We can add any two chains in an obvious way to obtain another chain by simply following the rule

$$n\sigma + m\sigma = (n + m)\sigma \quad (4.5)$$

Then the collection of all singular n -chains in X becomes a free abelian group, with the empty sum playing the role of the zero-element. The collection of all singular n -simplices is then a basis for this group. We denote this group by $S_n(X)$.

Proposition 6.1.2 ► Functoriality

If $f : X \rightarrow Y$ is a map of topological spaces, then $\sigma \mapsto f \circ \sigma$ extends to define a graded group homomorphism $f_* : S_n(X) \rightarrow S_n(Y)$. Since $(g \circ f) \circ \sigma = g \circ (f \circ \sigma)$, it follows that the association $X \mapsto S_n(X)$ defines a functor.

Definition 6.1.3 ► Relative Chain Complex

If A is a subspace of X and σ is a singular simplex in A then it can be considered as a singular simplex in X via the inclusion map $A \hookrightarrow X$. In this way, $S_n(A)$ becomes a subgroup of $S_n(X)$. We denote the quotient group $S_n(X)/S_n(A)$ by $S_n(X, A)$.

Remark.

1. $S_n(X, A)$ is a free abelian group with the collection of all singular n -simplexes σ in X whose image is not contained in A as a basis.

2. If $A = \emptyset$, then clearly $S_n(X, A) = S_n(X)$.

Definition 6.1.4

For each integer $r \geq 0$, the **face map** or the **face operator** $F^r : \mathbb{R}^\infty \rightarrow \mathbb{R}^\infty$ is defined by

$$F^r(\mathbf{e}_s) = \begin{cases} \mathbf{e}_s, & \text{if } s < r, \\ \mathbf{e}_{s+1}, & \text{if } s \geq r \end{cases}$$

and extended linearly.

Remark.

1. Note that for each $n \geq r$, F^r carries Δ_{n-1} onto the $(n-1)$ -face of Δ_n opposite to the vertex $\mathbf{e}_r$, in an **order preserving** manner.
2. We shall denote each F^r restricted to any $\Delta_n, n \geq r$ also by F^r , the exact meaning being understood from the context.

Lemma 6.1.5

$$F^r \circ F^{s-1} = F^s \circ F^r \text{ if } r < s. \text{ }^a$$

^a这里修正了 shastri 书中的一个笔误, 他可能写错了 $s-1$ 的位置

Proof. Since $r < s, r \leq s-1$, we fully expand the LHS by definition

$$\begin{aligned} F^r \circ F^{s-1}(\mathbf{e}_k) &= \begin{cases} F^r(\mathbf{e}_k), & k < s-1 \\ F^r(\mathbf{e}_{k+1}), & k \geq s-1 \end{cases} \\ &= \begin{cases} \mathbf{e}_k, & k < s-1, k < r \\ \mathbf{e}_{k+1}, & r \leq k < s-1 \\ \mathbf{e}_{k+1}, & s-1 \leq k < r-1 \text{ (empty set)} \\ \mathbf{e}_{k+2}, & k \geq s-1, k \geq r-1 \end{cases} \\ &= \begin{cases} \mathbf{e}_k, & k < r \\ \mathbf{e}_{k+1}, & r \leq k < s-1 \\ \mathbf{e}_{k+2}, & k \geq s-1 \end{cases} \end{aligned}$$

Similarly,

$$\begin{aligned}
 F^s \circ F^r (\mathbf{e}_k) &= \begin{cases} F^s (\mathbf{e}_k), & k < r \\ F^s (\mathbf{e}_{k+1}), & k \geq r \end{cases} \\
 &= \begin{cases} \mathbf{e}_k, & k < r, k \leq s \\ \mathbf{e}_{k+1}, & k < r, k \geq s \text{ (empty set)} \\ \mathbf{e}_{k+1}, & k \geq r, k < s-1 \\ \mathbf{e}_{k+2}, & k \geq r, k \geq s-1 \end{cases} \\
 &= \begin{cases} \mathbf{e}_k, & k < r \\ \mathbf{e}_{k+1}, & r \leq k < s-1 \\ \mathbf{e}_{k+2}, & k \geq s-1 \end{cases}
 \end{aligned}$$

□

Definition 6.1.6

For $n \geq 1$ and for any singular n -simplex σ in X , we define

$$\partial_n(\sigma) = \sum_{r=0}^n (-1)^r \sigma \circ F^r$$

and extend it linearly to obtain homomorphisms $\partial_n : S_n(X, A) \rightarrow S_{n-1}(X, A)$. We define $S_n(X, A) = (0)$ for $n \leq -1$ and $\partial_n = 0$ for $n \geq 0$.

Note.

σ is a map from Δ_n to X . $\sigma \circ F^r$ is the restriction of σ on its face opposite to $\mathbf{e}_r$. Then $\partial_n(\sigma)$ is something like a closed $(n-1)$ -surface consists of σ 's faces.

Proposition 6.1.7 ► Singular Chain Complex

$(S(X, A), \partial) = \{S_n(X, A), \partial_n\}$ is a chain complex and the association $(X, A) \rightsquigarrow (S(X, A), \partial)$ is a functor.

Proof. We shall first show that $\partial^2(\sigma) = 0$ for any singular n -simplex, $n \geq 2$:

$$\begin{aligned}
 \partial^2(\sigma) &= \partial \left(\sum_{s=0}^n (-1)^s \sigma \circ F^s \right) \\
 &= \sum_{s=0}^n (-1)^s \partial(\sigma \circ F^s) \\
 &= \sum_{s=0}^n \sum_{r=0}^{n-1} (-1)^s (-1)^r \sigma \circ F^s \circ F^r \\
 &= \sum_{0 \leq r < s \leq n} (-1)^{s+r} \sigma \circ F^r \circ F^{s-1} + \sum_{n-1 \geq r \geq s \geq 0} (-1)^{s+r} \sigma \circ F^s \circ F^r \\
 &= \sum_{0 \leq r < s \leq n} (-1)^{s+r} \sigma \circ F^r \circ F^{s-1} + \sum_{n-1 \geq s \geq r \geq 0} (-1)^{s+r} \sigma \circ F^r \circ F^s \\
 &= \sum_{0 \leq r < s \leq n} (-1)^{s+r} \sigma \circ F^r \circ F^{s-1} - \sum_{n \geq s > r \geq 0} (-1)^{s+r} \sigma \circ F^r \circ F^{s-1} \\
 &= 0
 \end{aligned}$$

Given a map $f : (X, A) \rightarrow (Y, B)$. Since $(f \circ \sigma) \circ F^r = f \circ (\sigma \circ F^r)$, we define $f_*(\sigma) := f \circ \sigma$ extended by linearity. Then $\partial(f_*(\sigma)) = f_*(\partial(\sigma))$ by the definition of ∂ and f_* 's linearity. The rest of the details are omitted. $\square$

Definition 6.1.8

The **relative singular homology groups** of a topological pair (X, A) are the homology groups $H_*(S_*(X, A))$ and are denoted by $H_*(X, A)$. Taking $A = \emptyset$, we get the homology groups $H_*(X)$.

Remark.

Given any commutative ring R

1. We could have formed the **free module** $S_n(X; R)$ of all the singular n -chains in X .
2. Exactly as before, we would have got a chain complex $S_*(X; R)$, sub-chain complex $S_*(A; R)$ and the quotient complex $S_*(X, A; R)$, etc.
3. The corresponding homology groups $H_*(X; R)$, etc., are then called the homology groups X with coefficients in R . Clearly they form a **graded R -module** by themselves. It is customary to drop the notation R when $R = \mathbb{Z}$.

6.2 Basic Properties

Remark.

- It is easily seen that $\{S_*(X, A), \partial\}$ is a functor from the category of topological pairs to the category of chain complexes; thus $H_*(X, A)$ and $H_*(X)$ are also functors.
- Given $f : (X, A) \rightarrow (Y, B)$, we shall denote by f_* , the map induced by f on $S_*(X, A)$ and by f_* the map induced by f on $H_*(X, A)$.

Lemma 6.2.1

There exist functorial chain homotopies

$$h : S_q(X, A) \rightarrow S_{q+1}((X, A) \times \mathbb{I}), q \geq 0,$$

,called **prism operator**, between the two inclusion induced maps $\eta_0, \eta_1 : S_*(X, A) \rightarrow S_*((X, A) \times \mathbb{I})$, where for any $t \in \mathbb{I}$, η_t is the inclusion map

$$\eta_t(x) = (x, t), \quad t \in \mathbb{I}$$

Theorem 6.2.2 ▶ Homotopy invariance

Let $f, g : (X, A) \rightarrow (Y, B)$ be **homotopic** to **each other**. Then $f_* = g_*$ on $H_*(X, A)$.

Proof. Let $G : (X, A) \times \mathbb{I} \rightarrow (Y, B)$ be a homotopy between f and g . For any $t \in \mathbb{I}$, let $\eta_t : (X, A) \rightarrow (X, A) \times \mathbb{I}$ be inclusion map $\eta_t(x) = (x, t)$. Then $G \circ \eta_0 = f$, $G \circ \eta_1 = g$. Only need to show that chain maps $(\eta_0)_*$ and $(\eta_1)_*$ are chain homotopic, thus $(\eta_0)_* = (\eta_1)_*$, $f_* = g_*$, which is implied by the lemma above. $\square$

Corollary 6.2.3

Homotopy-equivalent topological spaces have isomorphic homology groups; in particular, a contractible space has the homology groups of a point-space.

Example 6.2.4

Let $\star$ be a point space. We have

$$H_k(\star) = \begin{cases} \mathbb{Z}, & k = 0 \\ 0, & k \geq 1 \end{cases}$$

Proof. Let n be any positive integer. There is a unique singular n -simplex $\sigma : |\Delta_n| \rightarrow \{\star\}$ and hence $S_n(\star) \cong \mathbb{Z}$ for each $n \geq 0$. Observe that $\partial : S_{2n+1}(\star) \rightarrow S_{2n}(\star)$ is the zero-map because Δ_{2n+1} has an even number of $(2n)$ -faces and the terms in $\partial(\sigma)$ cancel in pairs. However, similar reasoning tells us that $\partial : S_{2n}(\star) \rightarrow S_{2n-1}(\star)$ is an isomorphism. Thus we have a chain complex

$$\dots \xrightarrow{0} \mathbb{Z} \xrightarrow{\cong} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{\cong} \mathbb{Z} \longrightarrow 0$$

Thus it follows that $H_0(\star) \cong \mathbb{Z}$ and $H_i(\star) = (0)$ for all $i > 0$. $\square$

Example 6.2.5

Let $\{X_j : j \in J\}$ be the set of path connected components of X so that $X = \bigsqcup_{j \in J} X_j$. Then

$$H_*(X) = \bigoplus_{j \in J} H_*(X_j)$$

Remark.

Thus in practice, we can often assume that the space itself is path connected, while dealing with its homology groups.

Proof. Since $|\Delta_n|$ is a path connected space, it follows that any singular n -simplex in X is contained completely in one of the path components X_j . Therefore, it follows that the entire chain complex $S_*(X)$ is the direct sum of the subchain complexes $S_*(X_j)$. Hence the homology is also a direct sum of $H_*(X_j)$, $j \in J$ by Theorem 5.3.3. $\square$

Theorem 6.2.6 ▶ Homology Long Exact Sequence of the pair (X, A)

By definition of $S_*(X, A)$, we have an exact sequence of chain complexes

$$0 \rightarrow S_*(A) \rightarrow S_*(X) \rightarrow S_*(X, A) \rightarrow 0.$$

By Theorem 6.8.5, this yields the long exact sequence of homology groups

$$\begin{cases} \dots \rightarrow H_i(A) \rightarrow H_i(X) \rightarrow H_i(X, A) \xrightarrow{\delta} H_{i-1}(A) \rightarrow \dots \\ \dots \rightarrow H_1(X, A) \xrightarrow{\delta} H_0(A) \rightarrow H_0(X) \rightarrow H_0(X, A), \end{cases} \quad (4.6)$$

which is functorial.

Remark.

This sequence comes handy in many computations as we shall see soon.

6.3 Homology Excision and Mayer-Vietoris

Lemma 6.3.1

Let $X = A \cup B$. Then the following statements are equivalent.

1. $S_*(A) + S_*(B) \rightarrow S_*(X)$ induces isomorphisms in homology.
2. $[S_*(A) + S_*(B)]/S_*(B) \rightarrow S_*(X)/S_*(B)$ induces isomorphisms in homology.
3. $[S_*(A) + S_*(B)]/S_*(A) \rightarrow S_*(X)/S_*(A)$ induces isomorphisms in homology.
4. $S_*(A)/S_*(A \cap B) \rightarrow S_*(X)/S_*(B)$ induces isomorphisms in homology.
5. $S_*(B)/S_*(A \cap B) \rightarrow S_*(X)/S_*(A)$ induces isomorphisms in homology.

Note.

1,2 等价的好处就在于, 它告诉我们, 我们所需要的同调群的同构对于一部分的信息是不敏感的。

Proof sketch.

对于 1,2 的等价性, 涉及四种同调群, 我们借助 B 的同调将它们相互联系. 用短正合列到长正合列的诱导行为将四种同调群两两联系, 并以 B 的同调为桥梁, 利用诱导行为的函子性将两组同调群联系起来. B 到自身的恒等给出一组同构, 配合

已知的一组同构可以利用 Five lemma 夹出另一组同构.
2,4 的等价性直接就是第二同构定理的结果, 本质上与同调关系不大.

Proof. The equivalence of (1) and (2) is a consequence of the Five lemma applied to the ladder of homology long exact sequences given by the ladder of short exact sequences

$$\begin{array}{ccccccc}
 0 & \longrightarrow & S.(B) & \longrightarrow & S.(A) + S.(B) & \longrightarrow & [S.(A) + S.(B)]/S.(B) \longrightarrow 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & S.(B) & \longrightarrow & S.(X) & \longrightarrow & S.(X)/S.(B) \longrightarrow 0.
 \end{array}$$

The equivalence of (2) and (4) follows from the commutative diagram below in which the horizontal arrow is the isomorphism given by the Noether's isomorphism theorem.

$$\begin{array}{ccc}
 S.(A)/S.(A \cap B) & \longrightarrow & [S.(A) + S.(B)]/S.(B) \\
 & \searrow & \downarrow \\
 & & S.(X)/S.(B)
 \end{array}$$

□

Definition 6.3.2

Let X be a topological space, A, B be two subspaces such that $X = A \cup B$. We then say that the inclusion map $(A, A \cap B) \hookrightarrow (X, B)$ an excision map. A pair $\{A, B\}$ of subspaces of a space X is called an **excisive couple** for singular homology, if the inclusion map

$$S.(A) + S.(B) \hookrightarrow S.(A \cup B)$$

induces isomorphisms in homology or if any of the other four equivalent conditions in the above lemma is satisfied.

Note.

excisive couple 是一对能代表 (X, A) 和 (X, B) 的同调性质的子集 (A, B) , 是 X 的一个良好分解. 它使得全空间 X 在切除 $X \setminus A$ 或 $X \setminus B$ 的切除行为上, 在同调的意义下是无损的.

Theorem 6.3.3 ▶ Homology excision

The inclusion map $(A, A \cap B) \hookrightarrow (A \cup B, B)$ of an excisive couple $\{A, B\}$ induces an isomorphism of homology groups $H_*(A, A \cap B) \rightarrow H_*(A \cup B, B)$.

Theorem 6.3.4

If $X = X_1 \cup X_2 = \text{int}_X(X_1) \cup \text{int}_X(X_2)$ then, $\{X_1, X_2\}$ is an excisive couple for the singular homology.

Note.

也就是说, 当分解的一个组件的边界能被另一个组件的内部“吃掉”时, 这组分解就是切除对.

Example 6.3.5

Here are a few examples of excisive couples which occur in practice.

1. $\{\mathbb{D}^n, \mathbb{R}^n \setminus \{0\}\}$ is an excisive couple and hence the inclusion map $(\mathbb{D}^n, \mathbb{D}^n \setminus \{0\}) \hookrightarrow (\mathbb{R}^n, \mathbb{R}^n \setminus \{0\})$ will induce isomorphisms in all singular homology groups.
2. Let N denote the north pole on $\mathbb{S}^n$ and $\bar{U}$ denote the closed upper hemisphere. Then $\{\mathbb{S}^n \setminus \{N\}, \bar{U}\}$ is an excisive couple, which gives isomorphisms of $H_*(\bar{U}, \bar{U} \setminus \{N\}) \rightarrow H_*(\mathbb{S}^n, \mathbb{S}^n \setminus \{N\})$.

Theorem 6.3.6

Given subsets X_1 and X_2 of a topological space X , let us denote the inclusion maps of X_i in to X by $\eta_i, i = 1, 2$. They induce inclusion maps of the chain complexes which we shall denote again by $\eta_i : S.(X_i) \rightarrow S.(X), i = 1, 2$. Now consider the chain map

$$(\eta_1, -\eta_2) : S.(X_1) \oplus S.(X_2) \rightarrow S.(X).$$

The submodule generated by $S.(X_1)$ and $S.(X_2)$ in $S.(X)$ is denoted by $S.(X_1) + S.(X_2)$. Then we have a short exact sequence of chain complexes

$$0 \rightarrow S.(X_1 \cap X_2) \rightarrow S.(X_1) \oplus S.(X_2) \rightarrow S.(X_1) + S.(X_2) \rightarrow 0$$

If $\{X_1, X_2\}$ is an excisive couple, then we can obtain the **Mayer-Vietoris**

sequence:

$$\left. \begin{aligned} \cdots \rightarrow H_{i+1}(X_1 \cup X_2) \xrightarrow{\partial} H_i(X_1 \cap X_2) \xrightarrow{i_*} H_i(X_1) \oplus H_i(X_2) \xrightarrow{j_*} H_i(X_1 \cup X_2) \rightarrow \cdots \\ \cdots \rightarrow H_1(X_1 \cup X_2) \xrightarrow{\partial} H_0(X_1 \cap X_2) \rightarrow H_0(X_1) \oplus H_0(X_2) \rightarrow H_0(X_1 \cup X_2) \end{aligned} \right\}$$

with

$$i_*(z) = (i_{1*}z, -i_{2*}z); \quad j_*(z_1, z_2) = j_{1*}z_1 + j_{2*}z_2$$

where,

$$\begin{aligned} i_1 : X_1 \cap X_2 \hookrightarrow X_1, \quad i_2 : X_1 \cap X_2 \hookrightarrow X_2 \\ j_1 : X_1 \hookrightarrow X_1 \cup X_2, \quad j_2 : X_2 \hookrightarrow X_1 \cup X_2 \end{aligned}$$

are inclusion maps.

Remark.

The Mayer-Vietoris sequence is one of the major ready-to-use-tools in computing homology.

Proof. The kernel of $(\eta_1, -\eta_2)$ is $S.(X_1) \cap S.(X_2) = S.(X_1 \cap X_2)$. The image is clearly $S.(X_1) + S.(X_2)$. Thus the short sequence is exact. Then we obtain the long exact sequence

$$\begin{aligned} \cdots \rightarrow H_{i+1}(S.(X_1) + S.(X_2)) \xrightarrow{\partial} H_i(X_1 \cap X_2) \xrightarrow{i_*} H_i(X_1) \oplus H_i(X_2) \xrightarrow{j_*} H_i(S.(X_1) + S.(X_2)) \\ \cdots \rightarrow H_1(S.(X_1) + S.(X_2)) \xrightarrow{\partial} H_0(X_1 \cap X_2) \rightarrow H_0(X_1) \oplus H_0(X_2) \rightarrow H_0(S.(X_1) + S.(X_2)) \end{aligned}$$

By the definition of excisive couple, we can replace $H_i(S.(X_1) + S.(X_2))$ by $H_i(X_1 \cup X_2)$ to obtain Mayer-Vietoris. $\square$

6.4 Problems

Problem 6.4.1

Compute the homology of $\mathbb{S}^p \times \mathbb{S}^q$ using Mayer-Vietoris.

Proof. We assume that $p, q \geq 1$. Let U^+, U^- be $\mathbb{S}^q$ without the north pole and south pole respectively. Then $(A, B) := (\mathbb{S}^p \times U^+, \mathbb{S}^p \times U^-)$ is a excisive

couple of $\mathbb{S}^p \times \mathbb{S}^q$. Since $U^+ \cap U^-$ can deformation retract to the equator of $\mathbb{S}^q$, which is $\mathbb{S}^{q-1}$, then $A \cap B$ can deformation retract to $\mathbb{S}^p \times \mathbb{S}^{q-1}$.

$$\cdots \xrightarrow{i'_*} H_{i+1}(\mathbb{S}^p) \oplus H_{i+1}(\mathbb{S}^q) \xrightarrow{j_*} H_{i+1}(\mathbb{S}^p \times \mathbb{S}^q) \xrightarrow{\partial} H_i(\mathbb{S}^p \times \mathbb{S}^{q-1}) \xrightarrow{i_*} H_i(\mathbb{S}^p) \oplus H_i(\mathbb{S}^q) \cdots$$

The following short exact sequence separates

$$0 \rightarrow \ker \partial \xrightarrow{\iota} H_{i+1}(\mathbb{S}^p \times \mathbb{S}^q) \xrightarrow{\partial} \text{Im } \partial \rightarrow 0$$

which is equivalent to

$$0 \rightarrow \text{Im } j_* \simeq \text{Coker } j_* \xrightarrow{\iota} H_{i+1}(\mathbb{S}^p \times \mathbb{S}^q) \xrightarrow{\partial} \ker i_* \rightarrow 0$$

where $\text{Im } j_* \simeq H_{i+1}(\mathbb{S}^p) \oplus H_{i+1}(\mathbb{S}^q) / \text{Im } i'_* = \text{Coker } i'_*$. Thus we have

$$H_{i+1}(\mathbb{S}^p \times \mathbb{S}^q) \simeq \ker i_* \oplus \text{Coker } i'_*$$

$\text{Im } i_* \simeq H_i$

1. When $p = q$

(a)

□

Exercise 6.4.2

Compute $H_*(\vee_k \mathbb{S}^{n_k})$ where $\vee_k \mathbb{S}^{n_k}$ denotes the bouquet of spheres of varying dimensions n_k . In particular, compute $H_*(\vee_k \mathbb{S}^n)$ where each $n_k = n$.

Proof. Let U be a sufficiently small open neighbourhood near the base point. Let $X_1 = \vee_{k-1} \mathbb{S}^{n_{k-1}} \cup U$. $X_2 = \mathbb{S}^{n_k} \cup U$. X_1 is homotopic equivalent to $\vee_{k-1} \mathbb{S}^{n_{k-1}}$. X_2 is homotopic equivalent to $\mathbb{S}^{n_k}$. $X_1 \cap X_2 = U$ is contractible. The Mayer-Vietoris sequence is

$$\cdots H_{l+1}(\vee_k \mathbb{S}^{n_k}) \xrightarrow{\partial} 0 \rightarrow H_l(\vee_{k-1} \mathbb{S}^{n_{k-1}}) \oplus H_l(\mathbb{S}^{n_k}) \rightarrow H_l(\vee_k \mathbb{S}^{n_k}) \xrightarrow{\partial} 0 \cdots$$

when $l \geq 2$. Thus

$$H_l(\vee_{k-1} \mathbb{S}^{n_{k-1}}) \oplus H_l(\mathbb{S}^{n_k}) \simeq H_l(\vee_k \mathbb{S}^{n_k}), \quad l \geq 2$$

By induction , we have

$$H_l(\vee_k \mathbb{S}^{n_k}) \simeq \bigoplus_k H_l(\mathbb{S}^{n_k}), \quad l \geq 2$$

The M-V sequence reduces to

$$0 \rightarrow H_1(\vee_{k-1} \mathbb{S}^{n_{k-1}}) \oplus H_1(\mathbb{S}^{n_k}) \rightarrow H_1(\vee_k \mathbb{S}^{n_k}) \xrightarrow{\partial} H_0(U) \rightarrow H_0(\vee_{k-1} \mathbb{S}^{n_{k-1}}) \oplus H_0(\mathbb{S}^{n_k}) \dots$$

when $l = 1$. The exactness show that

$$\text{Im } \partial = \ker(H_0(U) \rightarrow H_0(\vee_{k-1} \mathbb{S}^{n_{k-1}}) \oplus H_0(\mathbb{S}^{n_k})) = 0$$

Thus

$$0 \rightarrow H_1(\vee_{k-1} \mathbb{S}^{n_{k-1}}) \oplus H_1(\mathbb{S}^{n_k}) \rightarrow H_1(\vee_k \mathbb{S}^{n_k}) \rightarrow 0$$

And

$$H_1(\vee_k \mathbb{S}^{n_k}) \simeq H_1(\vee_{k-1} \mathbb{S}^{n_{k-1}}) \oplus H_1(\mathbb{S}^{n_k})$$

By induction

$$H_1(\vee_k \mathbb{S}^{n_k}) \simeq \bigoplus_k H_1(\mathbb{S}^{n_k})$$

And since $\vee_k \mathbb{S}^{n_k}$ is path-connected , finally we have

$$H_l(\vee_k \mathbb{S}^{n_k}) = \begin{cases} \mathbb{Z}, & l = 0 \\ \bigoplus_k H_l(\mathbb{S}^{n_k}), & l \geq 1 \end{cases}$$

Where $H_l(\mathbb{S}^{n_k})$ we have already known. □

6.5 Basic Topology

Problem 6.5.1 ► Convex Bodies

A convex body is a compact, convex subset $K \subseteq \mathbb{R}^n$ which contains 0 as an interior point and which is symmetric around the origin .

1. Prove that the following formula defines a norm on $\mathbb{R}^n$:

$$N(x) := \inf \left\{ \lambda > 0 : \frac{x}{\lambda} \in K \right\}$$

2. Prove that K is homeomorphic to $\mathbb{D}^n$ and that ∂K is homeomorphic to $\mathbb{S}^{n-1}$.

Note.

1. 若 K 是一个单位球. 则 $N(x) = \|x\|$.
2. 证明 $N(x)$ 是一个 norm, 需要证明: 1. 非负正定 2. 绝对一次齐次 3. 三角不等式.
3. 由于 φ 的形式足够简单, 可以通过直接构造逆映射证明双射.

Lemma 6.5.2

X is compact, Y is Hausdorff. If $\exists f : X \rightarrow Y$ is a continuous bijection, then X, Y are homeomorphic.

Proof. 1. It is obvious that $N(x) \geq 0$. If $N(x) = 0$, then

$$\inf\left\{\lambda > 0 : \frac{x}{\lambda} \in K\right\} = 0$$

which means that for all $\varepsilon > 0$, $\frac{x}{\varepsilon} \in K$. Since K is compact, $\text{dist}(K, 0) < \infty$. If $x \neq 0$, then $\|x\| > 0$. We take $\varepsilon_0 < \frac{1}{\text{dist}(K, 0) \cdot \|x\|}$. Then $\left\|\frac{x}{\varepsilon_0}\right\| > \text{dist}(K, 0)$. But $\frac{x}{\varepsilon_0} \in K$, which is a contradiction. For the second part, let $x \in \mathbb{R}^n, k \in \mathbb{R} \setminus \{0\}$. For all $\lambda > N(x)$,

$$\frac{x}{\lambda} \in K \implies \frac{kx}{k\lambda} \in K$$

, which shows that $N(kx) \leq kN(x)$. Replace k by $\frac{1}{k}$ and replace x by kx , we have $N(kx) \geq kN(x)$. Thus $N(kx) = kN(x)$. For the triangle inequality, we need to show that

$$\inf\left\{\lambda > 0 : \frac{x_1 + x_2}{\lambda} \in K\right\} \leq \inf\left\{\lambda > 0 : \frac{x_1}{\lambda} \in K\right\} + \inf\left\{\lambda > 0 : \frac{x_2}{\lambda} \in K\right\}$$

Let $\lambda_0 = N(x_1 + x_2)$. Only need to show that

$$\frac{x_1 + x_2}{N(x_1) + N(x_2)} \in K$$

For all $\delta > 0$, we have

$$\frac{x_1}{N(x_1) + \delta} \in K, \quad \frac{x_2}{N(x_2) + \delta} \in K$$

. Since K is convex, we have the convex combination

$$\frac{1}{N(x_1) + N(x_2) + 2\delta} \left((N(x_2) + \delta) \frac{x_2}{N(x_2) + \delta} + (N(x_1) + \delta) \frac{x_1}{N(x_1) + \delta} \right) \in K$$

That is

$$\frac{x_1 + x_2}{N(x_1) + N(x_2) + 2\delta} \in K$$

for all $\delta > 0$. Then

$$N(x_1 + x_2) \leq N(x_1) + N(x_2) + 2\delta, \quad \forall \delta > 0 \implies N(x_1 + x_2) \leq N(x_1) + N(x_2)$$

2. $x \in K^\circ \iff N(x) < 1$, $x \in \text{Ext}(K) \iff N(x) > 1$, $x \in \partial K \iff N(x) = 1$. We define a map $\varphi : \mathbb{R}^n \rightarrow \mathbb{R}^n$.

$$\varphi(x) := \begin{cases} N(x) \frac{x}{\|x\|}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

We have

$$\|\varphi(x)\| = N(x)$$

Then φ maps K° on to $(\mathbb{D}^n)^\circ$, and maps ∂K on to $\mathbb{S}^{n-1}$. To show φ is continuous, only need to show it is continuous at 0. In fact, $\|\lim_{x \rightarrow 0} \varphi(x)\| = \lim_{x \rightarrow 0} \|\varphi(x)\| = \lim_{x \rightarrow 0} N(x) = 0$ since N is a norm. We define $\psi : \mathbb{R}^n \rightarrow \mathbb{R}^n$:

$$\psi(x) := \begin{cases} \frac{\|x\|}{N(x)} x, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

Then $\varphi \circ \psi = \psi \circ \varphi = \text{Id}$. The same skills can be used to show ψ is continuous at 0 as well. Finally, by the lemma above, X, Y are homeomorphic.

□

Problem 6.5.3 ► Projective spaces

Let $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. The n -dimensional projective space $\mathbb{K}\mathbb{P}^n$ is the orbit space of $\mathbb{K}^{n+1} \setminus \{0\}$ under the action of $\mathbb{K}^*$ by rescaling.

1. Prove that $\mathbb{R}\mathbb{P}^n$ is homeomorphic to the orbit space $\mathbb{S}^n \setminus \{\pm 1\}$, where $\{\pm 1\}$ acts by multiplication on $\mathbb{S}^n \subseteq \mathbb{R}^{n+1}$.
2. Prove that $\mathbb{C}\mathbb{P}^n$ is homeomorphic to the orbit space $\mathbb{S}^{2n+1} \setminus \mathbb{S}^1$,

where $\mathbb{S}^1 = \{z \in \mathbb{C} : |z| = 1\}$ acts on $\mathbb{S}^{2n+1} \subseteq \mathbb{C}^{n+1}$ by multiplication.

3. Prove that $\mathbb{RP}^1$ is homeomorphic to $\mathbb{S}^1$ and that $\mathbb{CP}^1$ is homeomorphic to $\mathbb{S}^2$.

Proof. 1. Define

$$\pi : \mathbb{R}^{n+1} \setminus \{0\} \rightarrow \mathbb{RP}^n, \quad \pi(x) = [x]$$

and define

$$f : \mathbb{S}^n \rightarrow \mathbb{RP}^n = \pi|_{\mathbb{S}^n}$$

Then f is continuous. Take any $p \in \mathbb{RP}^n$, there exists $v \in \mathbb{R}^{n+1} \setminus \{0\}$, such that $p = [v]$. Let $x = \frac{v}{\|v\|}$, then $x \in \mathbb{S}^n$,

$$f(x) = [x] = [v] = p$$

thus f is surjective. If

$$f(x) = f(y)$$

, then $[x] = [y]$, there exists $\lambda \in \mathbb{R}^*$, such that $x = \lambda y$. Since

$$\|x\| = \|\lambda y\| = |\lambda| \|y\| \implies |\lambda| = 1$$

, $\lambda = \pm 1$, $x = y$ or $x = -y$. Thus $\text{Fiber}_p(f) = \text{Orbit}_x(\{\pm 1\})$. The universal property of quotient topology gives a continuous bijection

$$\tilde{f} : \mathbb{S}^n \setminus \{\pm 1\} \rightarrow \mathbb{RP}^n$$

Since $\mathbb{S}^n \setminus \{\pm 1\}$ is compact, $\mathbb{RP}^n$ is Hausdorff, $\tilde{f}$ is a homeomorphism.

2. Similarly, let $g : \mathbb{S}^{2n+1} \rightarrow \mathbb{CP}^n$, $g(z) = [z]$. Take any $p = [w] \in \mathbb{CP}^n$, where $w \in \mathbb{C}^{n+1} \setminus \{0\}$. Let $z = \frac{w}{\|w\|}$. Then

$$g(z) = [z] = [w] = p$$

g is surjective. If

$$g(z_1) = g(z_2)$$

, then there exists $\lambda \in \mathbb{C}^*$, such that $z_1 = \lambda z_2$, then

$$\|z_1\| = |\lambda| \|z_2\| \implies |\lambda| = 1$$

which gives that $\lambda \in \mathbb{S}^1$. Thus $\text{Fiber}_p(f) \simeq \text{Orbit}_x(\mathbb{S}^1)$. Thus

$$\mathbb{CP}^n \simeq \mathbb{S}^{2n+1}/\mathbb{S}^1$$

3. (a) Only need to show that $\mathbb{S}^1/\{\pm 1\} \simeq \mathbb{S}^1$,

$$h(z) = z^2$$

induces a continuous bijection $\tilde{h} : \mathbb{S}^1 \setminus \{\pm 1\} \rightarrow \mathbb{S}^1$. $\mathbb{S}^1 \setminus \{\pm 1\}$ is compact, $\mathbb{S}^1$ is Hausdorff, $\tilde{h}$ is a homeomorphism.

(b) Since $\mathbb{S}^3/\mathbb{S}^1 \simeq \mathbb{S}^2$.

□

Problem 6.5.4

The torus $\mathbb{T}$ is the quotient of $[0, 1]^2$ by the equivalence relation generated by $(x, 0) \sim (x, 1)$ and $(0, y) \sim (1, y)$ for all $x, y \in [0, 1]$. Prove that $\mathbb{T}$ is homeomorphic to:

1. The product $\mathbb{S}^1 \times \mathbb{S}^1$.
2. The quotient of $\mathbb{R}^2$ under the action of the (discrete) group $\mathbb{Z}^2$ by translations.

Proof. 1. Let

$$f(x, y) := (e^{2\pi xi}, e^{2\pi yi})$$

Then

$$f(x, 0) = (e^{2\pi ix}, 1) = f(x, 1)$$

$$f(0, y) = (1, e^{2\pi yi}) = f(1, y)$$

Thus f induces a map $\bar{f}$ on $\mathbb{T}$ such that $\bar{f}([(x, y)]) = (e^{2\pi xi}, e^{2\pi yi})$, which is a continuous function. It is not hard to see that $\bar{f} : \mathbb{T} \rightarrow \mathbb{S}^1 \times \mathbb{S}^1$ is a bijection. As a continuous image of $[0, 1]^2$, $\mathbb{T}$ is compact. Take any closed subset V of $\mathbb{T}$, V is compact. $\bar{f}(V)$ is compact as the continuous image of V . Since $\mathbb{S}^1 \times \mathbb{S}^1$ is Hausdorff, $\bar{f}(V)$ is closed. Thus $\bar{f}$ is a continuous and closed bijection, thus is a homeomorphism.

2. Let

$$f(x, y) = (x, y)/\mathbb{Z}^2$$

Then Since $(x, 0) - (x, 1) \in \mathbb{Z}^2, (0, y) - (1, y) \in \mathbb{Z}^2$. f induces a continuous map $\bar{f}$ on from $\mathbb{T}$ to $\mathbb{S}^1 \times \mathbb{S}^1$. It is clear that f maps onto $\mathbb{R}^2/\mathbb{Z}^2$. If

$f(x_1, y_1) = f(x_2, y_2)$, then

$$(x_1 - x_2, y_1 - y_2) \in \mathbb{Z}^2 \implies x_1 = x_2 \text{ or } \{x_1, x_2\} = \{0, 1\}, \text{ and } y_1 = y_2 \text{ or } \{y_1, y_2\} = \{0, 1\}$$

thus $(x_1, y_1) \sim (x_2, y_2)$. Again since $\mathbb{T}$ is compact, $\mathbb{R}^2/\mathbb{Z}^2$ is Hausdorff, $\bar{f}$ is a homomorphism. □

Problem 6.5.5

Consider the orthogonal group $O_{n-1}(\mathbb{R})$ as the subgroup of $O_n(\mathbb{R})$ of matrices of the form:

$$\begin{pmatrix} 1 & 0 \\ 0 & A \end{pmatrix} \in O_n(\mathbb{R}), A \in O_{n-1}(\mathbb{R}).$$

Let $O_{n-1}(\mathbb{R})$ act on $O_n(\mathbb{R})$ by left multiplication. Prove that the orbit space $O_n(\mathbb{R})/O_{n-1}(\mathbb{R})$ is homeomorphic to $\mathbb{S}^{n-1}$.

Proof sketch.

将 $O_n(\mathbb{R})$ 刻画为 $\mathbb{S}^{n-1}$ 上的群作用. 此时 $A \sim \begin{pmatrix} 1 & 0 \\ 0 & A \end{pmatrix}$ 就是北极点的稳定子群. 一但作用是传递的, 由流形上的轨道-稳定化子定理, 就有

$$O_n(\mathbb{R})/O_{n-1}(\mathbb{R}) \simeq \mathbb{S}^{n-1}$$

6.6 Homotopy

Problem 6.6.1 ▶ Möbius band

Let M be the Möbius band, that is, the quotient of $[0, 1]^2$ under the equivalence relation generated by $(x, 0) \sim (1 - x, 1)$ for all $x \in [0, 1]$. Prove that M is homotopy equivalent to $\mathbb{S}^1$.

Proof. Define a map

$$H : [0, 1]^2 \times [0, 1] \rightarrow \mathbb{S}^1, \quad H((x, y), t) = \left((1 - t)x + \frac{1}{2}t, y \right)$$

Let $\tilde{H} = \pi \circ H$. Then

$$\tilde{H}((x, 0), t) = \pi\left((1-t)x + \frac{1}{2}t, 0\right)$$

$$\tilde{H}((1-x, 1), t) = \pi\left(\left((1-t)(1-x) + \frac{1}{2}t\right), 1\right)$$

Since

$$1 - \left((1-t)x + \frac{1}{2}t\right) = (1-t)(1-x) + \frac{1}{2}t$$

,

$$\tilde{H}((x, 0), t) = \tilde{H}((1-x, 1), t)$$

$\tilde{H}$ induces a continuous map H^* on M . And

$$H^*([(x, y), 0]) = [(x, y)], \quad H^*([(x, y), 1]) = \left[\left(\frac{1}{2}, y\right)\right]$$

Thus H^* is a deformation retract from M to $\mathbb{S}^1$, which is a homotopy equivalence as well. $\square$

Problem 6.6.2

Let X be space. The cone CX is the quotient space:

$$CX = (X \times [0, 1]) / (X \times \{0\}).$$

Let $\pi : X \times [0, 1] \rightarrow CX$ be the quotient map and $\iota : X \rightarrow CX, x \mapsto \pi(x, 1)$.

1. Let $f : X \rightarrow Y$ be a map. Prove that f is homotopic to a constant map if and only if there exists $f' : CX \rightarrow Y$ such that $f' \circ \iota = f$.
2. Prove that CS^n is homeomorphic to D^{n+1} .
3. Prove that for all X , CX is contractible.

Proof. 1. If such f' exists. Then $H := f' \circ \pi : X \times [0, 1] \rightarrow Y$ is a homotopy from f to the constant map valued the vertex of CX . Conversely, if f is homotopic to constant map. Suppose $H : X \times I \rightarrow Y$ is its homotopy map. Since CX is the quotient space of $X \times [0, 1]$, and $H(X \times \{0\}) = \text{constant}$. It follows from the universal property that H induces a map $f' : CX \rightarrow Y$ such that $f' \circ \pi = H$. Finally

$$(f' \circ \iota)(x) = f'(\pi(x, 1)) = H(x, 1) = f(x)$$

2. Define

$$g : \mathbb{S}^n \times [0, 1] \rightarrow D^{n+1}, \quad g(x, t) = t \cdot x$$

Since $g(x, 0) \equiv 0 \in D^{n+1}$, g induces a continuous function $h : CS^n \rightarrow D^{n+1}$ such that $g \circ \pi = h$. We now show that h is a bijection. If

$$h([x_1, t_1]) = h([x_2, t_2])$$

, that is $t_1 x_1 = t_2 x_2$, then

$$\|t_1 x_1\| = \|t_2 x_2\| \implies t_1 = t_2$$

If $t_1 = t_2 = 0$, then $[x_1, t_1] = [x_2, t_2] = 0$. If $t_1 = t_2 > 0$, then $x_1 = x_2$, thus $[x_1, t_1] = [x_2, t_2]$. Thus h is injective. For surjectivity, let $y \in D^{n+1}$. If $y = 0$, we take $t = 0$, then $h([x, 0] = 0) = y$. If $y \neq 0$, then

$$h\left(\left[\frac{y}{\|y\|}, \|y\|\right]\right) = y$$

Thus h is surjective. Since CS^n is compact, D^{n+1} is Hausdorff, h is homeomorphism.

3. Define

$$H : CX \times [0, 1] \rightarrow CX, \quad H([x, s], t) = [x, s \cdot (1 - t)]$$

Then it is not hard to see that H is well-defined and continuous. Since

$$H([x, s], 0) = [x, s], \quad H([x, s], 1) = [x, 0]$$

H is a homotopy from Id to Constant.

□

Problem 6.6.3 ► Linear Groups

1. Prove that the inclusion $O_n(\mathbb{R}) \rightarrow GL_n(\mathbb{R})$ is a homotopy equivalence. (Use the Gram-Schmidt orthonormalization procedure to construct the reverse map).
2. Among the following matrix groups, determine which ones are compact, and determine their π_0 : $GL_n(\mathbb{C})$, $GL_n(\mathbb{R})$, $O_n(\mathbb{R})$, $SO_n(\mathbb{R})$, $U_n(\mathbb{C})$, $SU_n(\mathbb{C})$.

Proof. 1. Take $A \in GL_n(\mathbb{R})$, suppose $A = (v_1, \dots, v_n)$. Then $\{v_1, \dots, v_n\}$ passing to an orthonormal basis $\{u_1, \dots, u_n\}$ by Gram-Schmidt orthonor-

malization. We write $r(A) = \{u_1, \dots, u_n\}$. Then r is a continuous map on $GL_n(\mathbb{R})$ since the inner product is continuous. We define

$$H : GL_n(\mathbb{R}) \times [0, 1] \rightarrow GL_n(\mathbb{R}), \quad H(A, t) = (1 - t)A + tr(A)$$

we need to verify that $H(A, t) \in GL_n(\mathbb{R})$ for all A, t . The k -th column vector of $H(A, t)$ is $(1 - t)v_k + tu_k$. Since

$$v_k \in \text{span}\{u_1, \dots, u_{k-1}\}$$

we can write

$$(1 - t)v_k + tu_k = (1 - t) \sum_{j=1}^k c_{kj}u_j + tu_k$$

Since $c_{jj} > 0$, $(1 - t)c_{jj} + t > 0$, $\{(1 - t)v_1 + tu_1, \dots, (1 - t)v_k + tu_k\}$ are linealy independent. And since

$$H(A, 0) = A = \text{Id}_{GL_n(\mathbb{R})}(A)$$

$$H(A, 1) = r(A) \in O_n(\mathbb{R})$$

and for $B \in O_n(\mathbb{R})$, $H(B, t) = r(B) = 1$. We have $H : \text{Id}_{GL_n(\mathbb{R})} \simeq i$ is a SDR.

2. $GL_n(\mathbb{C})$ is a open submanifold of $M_n(\mathbb{C})$, which is noncompact. Since $GL_n(\mathbb{C})$ is path-connected. $\pi_0(GL_n(\mathbb{C})) = 1$. $GL_n(\mathbb{R})$ is a open submanifold of $M_n(\mathbb{R})$, thus $GL_n(\mathbb{R})$ is noncompact. $\det : GL_n(\mathbb{R}) \rightarrow \mathbb{R} \setminus \{0\}$ is a continuous surjection. Since $\mathbb{R} \setminus \{0\}$ has two components. $GL_n(\mathbb{R})$ has at least two components. Since

$$GL_n^+(\mathbb{R}) = \{A : \det A > 0\}, \quad GL_n^-(\mathbb{R}) = \{A : \det A < 0\}$$

are exactly two components of $GL_n(\mathbb{R})$. $\pi_0(GL_n(\mathbb{R})) = 2$. Since $\{A : A^T A = E\}$ is closed. $O_n(\mathbb{R})$ is bounded. $O_n(\mathbb{R})$ is compact. $\pi_0(O_n(\mathbb{R})) = 2$. $SO_n(\mathbb{R})$ is a closed subset of $O_n(\mathbb{R})$, thus is compact as well. $\pi_0(SO_n(\mathbb{R})) = 1$. $U_n(\mathbb{C})$ is a closed subset of $GL_n(\mathbb{C})$. $\pi_0(U_n(\mathbb{C})) = 1$. $SU_n(\mathbb{C})$ is compact, $\pi_0(SU_n(\mathbb{C})) = 1$.

□

Problem 6.6.4

1. Prove that $\mathbb{CP}^{n+1}$ is obtained from $\mathbb{CP}^n$ by gluing a cell of dimension $2n + 2$.
2. Compute the fundamental group of $\mathbb{CP}^n$ for $n \geq 1$.

6.7 van Kampen

Lemma 6.7.1

设 X 分解为一些道路连通集 A_α 的并, 且每个 A_α 包含了基点 $x_0 \in X$. 则所有含入映射 $A_\alpha \hookrightarrow X$ 的诱导同态 $j_\alpha : \pi_1(A_\alpha) \rightarrow \pi_1(X)$ 扩张为同态 $\Phi : *_\alpha \pi_1(A_\alpha) \rightarrow \pi_1(X)$.

Proposition 6.7.2

令 $i_{\alpha\beta} : \pi_1(A_\alpha \cap A_\beta) \rightarrow \pi_1(A_\alpha)$ 表示含入映射 $A_\alpha \cap A_\beta \hookrightarrow A_\alpha$ 诱导的同态, 则 $j_\alpha i_{\alpha\beta} = j_\beta i_{\beta\alpha}$. 所以 $\ker \Phi$ 包含所有形如 $i_{\alpha\beta}(\omega) i_{\beta\alpha}(\omega)^{-1}$, $\omega \in \pi_1(A_\alpha \cap A_\beta)$ 的元素.

Note.

即 Φ 将 $A_\alpha \cap A_\beta$ 落在 A_α 和 A_β 中部分的像等同.

Theorem 6.7.3 ▶ van Kampen

设 X 分解为包含了基点 $x_0 \in X$ 的道路连通开集 A_α 的并. 且每个交集 $A_\alpha \cap A_\beta$ 是道路连通的, 则同态 $\Phi : *_\alpha \pi_1(A_\alpha) \rightarrow \pi_1(X)$ 是满射. 此外, 若每个 $A_\alpha \cap A_\beta \cap A_\gamma$ 也是道路连通的, 则 Φ 的核 N 由全体形如 $i_{\alpha\beta}(\omega) i_{\beta\alpha}(\omega)^{-1}$, $\omega \in \pi_1(A_\alpha \cap A_\beta)$ 的元素生成, 从而 Φ 诱导出同构

$$\pi_1(X) \simeq *_\alpha \pi_1(A_\alpha) / N = *_\alpha \pi_1(A_\alpha) / (i_{\alpha\beta}(\omega) = i_{\beta\alpha}(\omega))$$

Example 6.7.4

设 $\{X_\alpha\}$ 是一族道路连通的空间, $x_\alpha \in X_\alpha$ 是基点, 考虑它们的楔和 $\bigvee_\alpha X_\alpha$. 对于每个 x_α , 它在 X_α 中都存在形变收缩到自身的开邻域 U_α . 令 $A_\alpha = X_\alpha \bigvee_{\beta \neq \alpha} U_\beta$, 则 A_α 形变收缩到 X_α . 此外, $\bigcap_\alpha A_\alpha$ 形变收缩到点 x_α . 由 van Kampen 定理, $\Phi : \pi_1(X) \rightarrow *_\alpha \pi_1(A_\alpha) \simeq *_\alpha \pi_1(X_\alpha)$ 是同构.

6.8 Homology

Lemma 6.8.1 ▶ Snake lemma

Given a commutative diagram of R -module homomorphisms: where the two horizontal sequences are exact, there exists a R -module homomorphism

$$\begin{array}{ccccccc} M' & \xrightarrow{\alpha} & M & \xrightarrow{\beta} & M'' & \longrightarrow & 0 \\ \downarrow f' & & \downarrow f & & \downarrow f'' & & \\ 0 & \longrightarrow & N' & \xrightarrow{\alpha'} & N & \xrightarrow{\beta'} & N'' \end{array}$$

$\delta : \text{Ker } f'' \longrightarrow \text{Coker } f'$, called the *connecting homomorphism* such that the sequence

$$\text{Ker } f' \xrightarrow{\bar{\alpha}} \text{Ker } f \xrightarrow{\bar{\beta}} \text{Ker } f'' \xrightarrow{\delta} \text{Coker } f' \xrightarrow{\bar{\alpha}'} \text{Coker } f \xrightarrow{\bar{\beta}'} \text{Coker } f''$$

is exact. Moreover, the *connecting homomorphism* δ has the naturality properties, so that the above assignment of a ‘snake’ to the corresponding ‘six-term’ exact sequence of modules defines a covariant functor.

Note.

最关键的信息是 *connecting homomorphism* δ . 构造的过程就是借助 β 的是 surjective, α' 是 injective, 来调整使得 β 和 α' 对应的箭头是可以反转的. 事实上在应用时, 这个 connecting homomorphism 的具体构造往往比它的存在性更重要.

Proof. to be finished We need to show

1. $\bar{\alpha}, \bar{\beta}$ are well defined, where $\bar{\alpha} := \alpha|_{\text{Ker } f'}, \bar{\beta} := \beta|_{\text{Ker } f}$
2. $\text{Im } \bar{\alpha} = \text{Ker } \bar{\beta}$.
3. $\bar{\alpha}', \bar{\beta}'$ are well defined, where $\bar{\alpha}'([x]) := [\alpha'(x)], \bar{\beta}'([y]) := [\beta'(y)]$
4. $\text{Im } \bar{\alpha}' = \text{Ker } \bar{\beta}'$
5. Construct a δ .
6. $\text{Im } \bar{\beta} = \text{Ker } \delta$

7. $\text{Im } \delta = \ker \bar{\alpha}'$

We prove the above points below.

1. By commutative, $f \circ \alpha(\ker f') = \alpha' \circ f'(\ker f') = 0$, which shows that $\alpha(\ker f') \subseteq \ker f$. Thus $\bar{\alpha}$ is well defined. $\bar{\beta}$'s is similar.

- 2.

$$\text{Im } \bar{\alpha} = \alpha(\ker f') \subseteq \text{Im } \alpha = \ker \beta \implies \beta(\text{Im } \bar{\alpha}) = 0 \implies \text{Im } \bar{\alpha} \subseteq \ker \bar{\beta}$$

The other side,

$$\ker \bar{\beta} = \ker \beta \cap \ker f = \text{Im } \alpha \cap \ker f$$

Take $y \in \ker \bar{\beta}$, there exists $x \in M'$ such that $y = \alpha(x)$ and $f(y) = f(\alpha(x)) = 0$. Since $f \circ \alpha = \alpha' \circ f'$, $\alpha' \circ f'(x) = 0$. Since α' is injective, $x \in \ker f'$. Thus $y \in \text{Im } \bar{\alpha}$.

3. For $\bar{\alpha}'$, we need to show that for $x_1, x_2 \in N'$ such that $x_1 - x_2 \in \text{Im } f'$, $\alpha'(x_1) - \alpha'(x_2) \in \text{Im } f$. It is true by commutative

$$\alpha'(x_1) - \alpha'(x_2) = \alpha'(x_1 - x_2) \in \text{Im } (\alpha' \circ f') = \text{Im } (f \circ \alpha) \subseteq \text{Im } f$$

$\bar{\beta}'$'s is similar.

4. Take $[y]_{\sim \text{Im } f} \in \text{Im } \bar{\alpha}'$, then there exists $[x]_{\sim \text{Im } f'} \in \text{Coker } f'$, such that $\alpha'([x]_{\sim \text{Im } f'}) = [y]_{\sim \text{Im } f}$. Then $\alpha'(x) - y \in \text{Im } f$, there exists $z \in M$, such that

$$\alpha'(x) - y = f(z)$$

By commutative and exactness: $\ker \beta' = \text{Im } \alpha'$,

$$\beta'(y) = \beta'(\alpha'(x) - f(z)) = 0 - \beta' \circ f(z) = -f'' \circ \beta(z) \in \text{Im } f''$$

, which shows that $\bar{\beta}'([y]_{\sim \text{Im } f}) = [0]_{\sim \text{Im } f''}$, $\text{Im } \bar{\alpha}' \subseteq \ker \bar{\beta}'$.

Then, take $[y]_{\sim \text{Im } f} \in \ker \bar{\beta}'$. Then $y \in \ker \beta' = \text{Im } \alpha'$. There exists $x \in N'$ such that $y = \alpha'(x)$. Then $[y]_{\sim \text{Im } f} = \alpha'([x]_{\sim \text{Im } f'})$, $[y]_{\sim \text{Im } f} \in \text{Im } \bar{\alpha}'$, $\ker \bar{\beta}' \subseteq \text{Im } \bar{\alpha}'$.

5. Take $z \in \ker f''$. Since β is surjective, there exists $y \in M$ such that $\beta(y) = z$. Once we show that $f(y) \in \text{Im } \alpha'$, we can define

$$\delta(z) = [(\alpha')^{-1}(f(y))]_{\sim \text{Im } f'}$$

provided that this definition is well defined. By commutative

$$\beta'(f(y)) = f''(\beta(y)) = f''(z) = 0.$$

Hence $f(y) \in \ker \beta' = \text{Im } \alpha'$. Finally, to show that it is well defined, we take $z_1, z_2 \in \ker f''$ such that $\beta(y_1) = \beta(y_2) = z$. We need to show that $(\alpha')^{-1}(f(y_1 - y_2)) \in \text{Im } f'$. Since $\beta(y_1) - \beta(y_2) = 0$, $\beta(y_1 - y_2) = 0$, $y_1 - y_2 \in \ker \beta = \text{Im } \alpha$. Suppose $\alpha(x) = y_1 - y_2$. Then

$$(\alpha')^{-1}(f(y_1 - y_2)) = (\alpha')^{-1}(f\alpha(x)) = (\alpha')^{-1}(\alpha' \circ f')(x) = f'(x) \in \text{Im } f'$$

, which completes the proof. □

Corollary 6.8.2

Consider the following commutative diagram of R -modules and R -linear maps in which the two rows are exact. If f_1 and f_3 are isomorphisms then so is f_2 .

$$\begin{array}{ccccccccc} 0 & \longrightarrow & M_1 & \xrightarrow{\alpha_1} & M_2 & \xrightarrow{\alpha_2} & M_3 & \longrightarrow & 0 \\ & & \downarrow f_1 & & \downarrow f_2 & & \downarrow f_3 & & \\ 0 & \longrightarrow & N_1 & \xrightarrow{\beta_1} & N_2 & \xrightarrow{\beta_2} & N_3 & \longrightarrow & 0 \end{array}$$

Proof. By snake lemma, we have

$$0 \xrightarrow{\alpha_1} \ker f_2 \xrightarrow{\alpha_2} 0 \xrightarrow{\delta} 0 \xrightarrow{\beta_1} \text{Coker } f_2 \xrightarrow{\beta_2} 0$$

is exact. Then we have

$$\begin{aligned} 0 &= \ker \alpha_2 = \text{Im } \alpha_1 = 0 \\ &= \ker f_2 \end{aligned}$$

which shows that $\ker f_2 = 0$. Similarly, $\text{Coker } f_2 = 0$. The above shows that f_2 is an isomorphism. □

Corollary 6.8.3 ▶ Four lemma

Consider the following commutative diagram of R modules and R -linear maps in which the two rows are exact. Suppose that f_1 is surjective and

f_4 is injective. Then

(i) f_2 is injective $\implies f_3$ is injective.

(ii) f_3 is surjective $\implies f_2$ is surjective.

$$\begin{array}{ccccccc} M_1 & \xrightarrow{\alpha_1} & M_2 & \xrightarrow{\alpha_2} & M_3 & \xrightarrow{\alpha_3} & M_4 \\ \downarrow f_1 & & \downarrow f_2 & & \downarrow f_3 & & \downarrow f_4 \\ N_1 & \xrightarrow{\beta_1} & N_2 & \xrightarrow{\beta_2} & N_3 & \xrightarrow{\beta_3} & N_4 \end{array}$$

Proof sketch.

如果 f_3 被两个单射 f_2, f_4 夹在中间, 得益于 f_1 是满射, 可以将右三列保持单射性地调整为一条 snake, 从而导出 f_3 是单射.

类似地, 如果 f_2 被两个满射 f_1, f_3 夹在中间, 得益于 f_4 是单射, 左三列可以保持满射性地调整为一条 snake, 导出 f_2 是满射.

Proof. 1. We can construct a snake

$$\begin{array}{ccccccc} M_2/\ker \alpha_2 & \xrightarrow{\alpha_2} & M_3 & \xrightarrow{\alpha_3} & \operatorname{Im} \alpha_3 & \longrightarrow & 0 \\ \downarrow \bar{f}_2 & & \downarrow f_3 & & \downarrow f_4|_{\operatorname{Im} \alpha_3} & & \\ 0 \longrightarrow & N_2/\ker \beta_2 & \xrightarrow{\beta_2} & N_3 & \xrightarrow{\beta_3} & N_4 & \end{array}$$

By snake lemma, the following sequence is exact

$$\ker \bar{f}_2 \rightarrow \ker f_3 \rightarrow \ker f_4|_{\operatorname{Im} \alpha_3}$$

Since f_4 is injective, $f_4|_{\operatorname{Im} \alpha_3}$ is as well. Furthermore, by applying the snake lemma to the following diagram

$$\begin{array}{ccccccc} M_1 & \xrightarrow{\alpha_1} & M_2 & \xrightarrow{p} & M_2/\ker \alpha_2 & \longrightarrow & 0 \\ \bar{f}_1 \downarrow & & \downarrow f_2 & & \downarrow \bar{f}_2 & & \\ 0 \longrightarrow & N_1/\ker \beta_1 & \xrightarrow{\bar{\beta}_1} & N_2 & \xrightarrow{p} & N_2/\ker \beta_2 & \end{array}$$

We get

$$0 = \ker f_2 \rightarrow \ker \bar{f}_2 \xrightarrow{\delta} \operatorname{Coker} \bar{f}_1$$

is a exact sequence, where $\operatorname{Coker} \bar{f}_1 = 0$ since f_1 is surjective and $\bar{f}_1$ is as

well. It shows that $\ker \bar{f}_2 = 0$. Finally, we have the following sequence is exact

$$0 = \ker \bar{f}_2 \rightarrow \ker f_3 \rightarrow \ker f_4|_{\text{Im } \alpha_3} = 0$$

It follows that $\ker f_3 = 0$, f_3 is injective.

2. Another snake we can construct is

$$\begin{array}{ccccccc} M_1 & \xrightarrow{\alpha_1} & M_2 & \xrightarrow{\alpha_2} & \text{Im } \alpha_2 = \ker \alpha_3 & \longrightarrow & 0 \\ \downarrow \bar{f}_1 & & \downarrow f_2 & & \downarrow f_3|_{\text{Im } \alpha_2} & & \\ 0 & \longrightarrow & N_1 / \ker \beta_1 & \xrightarrow{\bar{\beta}_1} & N_2 & \xrightarrow{\beta_2} & \text{Im } \beta_2 = \ker \beta_3 \end{array}$$

Then by snake lemma, the following sequence is exact

$$\ker \bar{f}_1 \rightarrow \ker f_2 \rightarrow \ker f_3|_{\text{Im } \alpha_2} \rightarrow \text{Coker } \bar{f}_1 \rightarrow \text{Coker } f_2 \rightarrow \text{Coker } f_3|_{\text{Im } \alpha_2}$$

Furthermore, by applying the snake lemma to the following diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & \text{Ker } \alpha_3 & \xrightarrow{\iota_M} & M_3 & \xrightarrow{\alpha_3} & \text{Im } \alpha_3 \longrightarrow 0 \\ & & \downarrow f_3|_{\text{Im } \alpha_2} & & \downarrow f_3 & & \downarrow f_4|_{\text{Im } \alpha_3} \\ 0 & \longrightarrow & \text{Ker } \beta_3 & \xrightarrow{\iota_N} & N_3 & \xrightarrow{\beta_3} & \text{Im } \beta_3 \longrightarrow 0 \end{array}$$

We get

$$0 = \ker f_4|_{\text{Im } \alpha_3} \xrightarrow{\delta} \text{Coker } (f_3|_{\text{Im } \alpha_2}) \rightarrow \text{Coker } f_3 = 0$$

which shows that $\text{Coker } (f_3|_{\text{Im } \alpha_2}) = 0$. Finally, we have the following sequence exact

$$0 = \text{Coker } \bar{f}_1 \rightarrow \text{Coker } f_2 \rightarrow \text{Coker } f_3|_{\text{Im } \alpha_2} = 0$$

Thus $\text{Coker } f_2 = 0$, f_2 is surjective. □

Corollary 6.8.4 ► Five lemma

In the following diagram of R -modules, the two rows are given to be exact. If f_1, f_2, f_4 and f_5 are isomorphisms then f_3 is also an isomorphism.

$$\begin{array}{ccccccccc}
M_1 & \longrightarrow & M_2 & \longrightarrow & M_3 & \longrightarrow & M_4 & \longrightarrow & M_5 \\
\downarrow f_1 & & \downarrow f_2 & & \downarrow f_3 & & \downarrow f_4 & & \downarrow f_5 \\
M'_1 & \longrightarrow & M'_2 & \longrightarrow & M'_3 & \longrightarrow & M'_4 & \longrightarrow & M'_5
\end{array}$$

Proof. By applying the Four lemma to the left four columns, we get f_3 is injective. And by applying the Four lemma to the right four columns, we get f_3 is surjective.

□

Theorem 6.8.5

Given a short exact sequence of chain complexes

$$0 \rightarrow C'_* \xrightarrow{\alpha} C_* \xrightarrow{\beta} C''_* \rightarrow 0$$

there is a functorial long exact sequence of homology groups

$$\rightarrow H_n(C'_*) \xrightarrow{H_n(\alpha)} H_n(C_*) \xrightarrow{H_n(\beta)} H_n(C''_*) \xrightarrow{\delta_n} H_{n-1}(C'_*) \xrightarrow{H_{n-1}(\alpha)} H_{n-1}(C_*) \rightarrow$$

Proof sketch.

Consider the diagram

$$\begin{array}{ccccccc}
C'_n/\text{Im}\partial'_{n+1} & \xrightarrow{\bar{\alpha}_n} & C_n/\text{Im}\partial_{n+1} & \xrightarrow{\bar{\beta}_n} & C''_n/\text{Im}\partial''_{n+1} & \longrightarrow & 0 \\
\downarrow \partial'_n & & \downarrow \partial_n & & \downarrow \partial''_{n+1} & & \\
0 \longrightarrow & \text{Ker}\partial'_{n-1} & \xrightarrow{\alpha'_{n-1}} & \text{Ker}\partial_{n-1} & \xrightarrow{\beta'_{n-1}} & \text{Ker}\partial''_{n-1} &
\end{array}$$

Example 6.8.6 ► reduced homology

定义 ε 为将 0-链映为系数和的同态. 称同态 ε 为增广同态, 可以按以下方式扩张

出链复形 $\tilde{S}(X)$

$$\tilde{S}_n(X) = \begin{cases} S_n(X), & \text{if } n \geq 0, \\ \mathbb{Z}, & \text{if } n = -1, \\ (0), & \text{if } n < -1; \end{cases} \quad \text{and} \quad \tilde{\partial}_n = \begin{cases} \partial_n, & n \geq 1, \\ \varepsilon, & n = 0, \\ 0, & n < 0. \end{cases}$$

相应的同调群记作 $\tilde{H}_*(X)$, 称为约化同调群. 显然 $\tilde{H}_i(X) = (0), i < 0$; $\tilde{H}_i(X) \simeq H_i(X), i > 1$; $\tilde{H}_0(X) \oplus \mathbb{Z} \simeq H_0(X)$. 特别地, 对于道路连通空间 X , $\tilde{H}_0(X) = 0$.

Problem 6.8.7 ► cofibration

Let (X, A) be a pair of spaces. We say that the inclusion $A \rightarrow X$ is a cofibration if, whenever $f : X \rightarrow Y$ and $H : A \times [0, 1] \rightarrow Y$ are maps such that $H(a, 0) = f(a)$ for all $a \in A$, there exists $\tilde{H} : X \times [0, 1] \rightarrow Y$ such that $\tilde{H}(a, t) = H(a, t)$ for all $(a, t) \in A \times [0, 1]$ and $\tilde{H}(x, 0) = f(x)$ for all $x \in X$.

1. Prove that the inclusion $(X, A) \rightarrow (X \cup CA, CA)$ induces an isomorphism on relative homology.
2. Prove that if $A \rightarrow X$ is a cofibration and A is contractible, then the quotient map $X \rightarrow X/A$ is a homotopy equivalence.
3. Prove that if X is obtained from A by gluing cells, then $A \rightarrow X$ is a cofibration.
4. Prove that if $A \rightarrow X$ is a cofibration, then $(X \cup CA, CA)$ is a cofibration.
5. Let $A \rightarrow X$ be a cofibration. Prove that the quotient map induces an isomorphism $H_i(X, A) \cong H_i(X/A, A/A) = \tilde{H}_i(X/A)$.

Problem 6.8.8 ► homology of suspension

Let X be a space. Compute $H_*(\Sigma X)$ in terms of $H_*(X)$.

Problem 6.8.9

Compute the homology of $\mathbb{CP}^n$.

6.9 Degree

Problem 6.9.1

Let $f : S^n \rightarrow S^n$ be a map. Its **degree** $\deg(f) \in \mathbb{Z}$ is the integer such that for all $x \in H_n(S^n; \mathbb{Z})$, we have $f_*(x) = \deg(f) \cdot x$.

1. Prove that this definition matches with the definition with fundamental groups for $n = 1$.
2. Prove that if $\deg(f) \neq 0$ then f is surjective. Find a counterexample for the converse.
3. Prove that if f is injective, then $\deg(f) = \pm 1$. Find a counterexample for the converse.
4. Prove that the degree of a reflection is -1 (use Mayer-Vietoris). Given $A \in O_n(\mathbb{R})$, prove that the degree of $x \mapsto Ax$ is equal to $\det(A)$. Compute the degree of the antipodal map $x \mapsto -x$.
5. Prove that ΣS^n is homeomorphic to S^{n+1} , then prove that the degree of f is equal to the degree of its suspension $\Sigma f : S^{n+1} \rightarrow S^{n+1}$.

Problem 6.9.2

1. Prove that any map $f : S^n \rightarrow S^n$ without fixpoints is homotopic to the antipodal map. (Use that for all $x \in S^n$, the origin $0 \in \mathbb{R}^{n+1}$ is not on the segment $[f(x), -x]$).
2. Prove that if n is even, then any $f : S^n \rightarrow S^n$ admits a fixpoint.
3. Prove that if a group G acts freely on S^n with n even, then $\#G \leq 2$.
4. Prove that S^n , for n odd, admits a nonvanishing vector field.
5. Prove that any vector field on S^n , for n even, vanishes at some point. (Given such a vector field X , consider the map $u \mapsto X(u)/\|X(u)\|$).

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